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Passive pedal force emulator assemblies

Abstract

Embodiments herein are directed to a pedal emulator assembly including a housing having a cavity defined by a pair of sidewalls, a first end wall and a second end wall. A lower arm having a spring retaining portion is formed therein. A pedal arm is pivotally coupled the pedal arm and includes a pedal pad. A lever arm including a spring receiving portion is positioned within the cavity of the housing and is pivotally coupled to the housing. A spring carrier including a compressible member extends between the spring receiving portion and the spring retaining portion. When a first predetermined load is applied to the pedal pad, the pedal arm drives the lever arm into the compressible member and the compressible member compresses into an at least partially compressed state to generate a first return force on the pedal pad.

Inventors: Kauz; Wjatscheslaw (Pattensen, DE), Leem; Daniel (Chatham, CA), Shon; Chong W. (Komoka, CA), Vanek; Marty (Thamesville, CA)

Applicant: KSR IP Holdings, LLC (Wilmington, DE)

Family ID: 90061265

Assignee: KSR IP Holdings, LLC (Wilmington, DE)

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Primary Examiner: Diaz; Thomas C

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This utility patent application claims priority benefit from U.S. provisional patent application Ser. No. 63/403,360, filed Sep. 2, 2022 and entitled “Passive Pedal Force Emulator Assemblies Having Friction and/or Lever Arms”, the entire contents of which is incorporated herein in its entirety.

TECHNICAL FIELD

(1) The present specification generally relates to pedal assemblies for vehicles and, more specifically, to passive force emulators for pedal assemblies with a mechanical resistance for damping and hysteresis dependent on a pedal movement.

BACKGROUND

(2) Traditional vehicle pedal systems are utilized in vehicles to perform functions such as acceleration, braking, and clutch operations. Typically, these pedals are mechanically linked to the respective vehicle components they control, resulting in a driver of the vehicle feeling direct feedback as they depress the pedal utilized in the pedal system. For example, a brake pedal may provide resistance as it pushes against the hydraulic fluid in a brake system. However, in many modern vehicles, such as electric and other autonomous vehicles, the direct mechanical linkages in pedal systems may be replaced with electronic systems. While these electronic systems may offer easier integration, fewer mechanical parts, and weight savings, they often lack the tactile feedback expected and relied upon by drivers for vehicle control. Accordingly, a need exists for a pedal emulator assembly that can recreate the feel and feedback of a traditional mechanical pedal.

SUMMARY

(3) In embodiments, a pedal emulator assembly is disclosed. The pedal emulator assembly includes

a housing having a cavity defined by a pair of sidewalls, a first end wall and a second end wall. A lower arm having a spring retaining portion is formed therein. A pedal arm is pivotally coupled to the lower arm and includes a pedal pad. A lever arm including a spring receiving portion is positioned within the cavity of the housing and is pivotally coupled to the housing. A spring carrier including a compressible member extends between the spring receiving portion and the spring retaining portion. When a first predetermined load is applied to the pedal pad, the pedal arm drives the lever arm into the compressible member and the compressible member compresses into an at least partially compressed state to generate a first return force on the pedal pad.

(4) In another embodiment, a pedal emulator assembly for a vehicle is disclosed. The pedal emulator assembly includes a floor-mounted housing having a cavity defined by a pair of sidewalls, a first end wall, and a second end wall. A lower arm having a spring retaining portion formed therein is positioned within the cavity. A pedal arm is at least partially received in the cavity, the pedal arm having an interior surface, a pivot end, and an opposite pad end on which a pedal pad is attached. A lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further including a first end, a second end positioned opposite the first end, an apex disposed at the first end, a hub portion disposed at the second end, an upper frame portion and a lower frame portion that connect at the apex and the hub portion; and a spring receiving portion positioned between the first end and the second end and within the lower frame portion. A spring carrier including a compressible member extends between the spring receiving portion and the spring retaining portion. When a first predetermined load is applied to the pedal pad, the interior surface of the pedal arm drives the lever arm into the compressible member such that the compressible member compresses into a compressed state to generate a first return force on the pedal pad.

(5) In yet another embodiment, a pedal emulator assembly for a vehicle is disclosed. The pedal emulator assembly includes a housing having a cavity defined by a pair of sidewalls, a first end wall and a second end wall. The pedal emulator assembly further includes a lower arm having a spring retaining portion formed therein. A pedal arm is at least partially received in the cavity, the pedal arm having a pivot end pivotally coupled to the housing, an opposite pad end on which a pedal pad is attached, and an interior surface that is non-linear. A lever arm is positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further includes a first end, a second end positioned opposite the first end, an apex disposed at the first end, a hub portion disposed at the second end, an upper frame and a lower frame that connect at the apex and the hub portion, the upper frame including an upper surface with a non-linear profile, and a spring receiving portion positioned between the first end and the second end in the lower frame. The pedal emulator assembly further includes a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion. When a first predetermined load is applied to the pedal pad, the interior surface of the pedal arm contacts the upper surface of the upper frame of the lever arm, such that the lever arm is forced into the compressible member to generate a return force on the pedal pad.

(6) These and additional features provided by the embodiments described herein will be more fully understood in view of the following detailed description, in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the subject matter defined by the claims. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

- (2) FIG. 1 schematically illustrates an example graphical representation of a pedal travel versus an applied force according to one or more embodiments shown and described herein;
- (3) FIG. 2 schematically illustrates an example graphical representation of a pedal travel versus a hysteresis force according to one or more embodiments shown and described herein;
- (4) FIG. 3A schematically depicts a left side perspective view of a first example pedal assembly, according to one or more embodiments shown and described herein;
- (5) FIG. 3B schematically depicts a rear and left side perspective view of the pedal assembly of FIG. 3A, according to one or more embodiments shown and described herein;
- (6) FIG. 3C schematically depicts an exploded perspective view of the pedal assembly of FIG. 3A, according to one or more embodiments shown and described herein;
- (7) FIG. 3D schematically depicts a cross-sectional view of the pedal assembly of FIG. 3A taken from line 3-3 with a pedal arm in an undepressed state, according to one or more embodiments shown and described herein;
- (8) FIG. 3E schematically depicts a cross-sectional view of the pedal assembly of FIG. 3A taken from line 3-3 with a pedal arm in an partially depressed state, according to one or more embodiments shown and described herein;
- (9) FIG. 3F schematically depicts a cross-sectional view of the pedal assembly of FIG. 3A taken from line 3-3 with a pedal arm in a fully depressed state, according to one or more embodiments shown and described herein;
- (10) FIG. 4A schematically depicts a left side perspective view of a second example pedal assembly, according to one or more embodiments shown and described herein;
- (11) FIG. 4B schematically depicts a rear and left side perspective view of the pedal assembly of FIG. 4A, according to one or more embodiments shown and described herein;
- (12) FIG. 4C schematically depicts an exploded perspective view of the pedal assembly of FIG. 4A, according to one or more embodiments shown and described herein;
- (13) FIG. 4D schematically depicts a cross-sectional view of the pedal assembly of FIG. 4A taken from line 4-4 with a pedal arm in an undepressed state, according to one or more embodiments shown and described herein;
- (14) FIG. 4E schematically depicts a cross-sectional view of the hanging pedal assembly of FIG. 4A taken from line 4-4 with a pedal arm in an partially depressed state, according to one or more embodiments shown and described herein;
- (15) FIG. 4F schematically depicts a cross-sectional view of the hanging pedal assembly of FIG. 4A taken from line 4-4 with a pedal arm in a fully depressed state, according to one or more embodiments shown and described herein;
- (16) FIG. 5A schematically depicts a left side perspective view of a third example pedal assembly utilizing friction within a slot to generate resistance, according to one or more embodiments shown and described herein;
- (17) FIG. 5B schematically depicts the left side perspective view of the pedal assembly of FIG. 5A without a housing, according to one or more embodiments shown and described herein;
- (18) FIG. 5C schematically depicts an exploded perspective view of the pedal assembly of FIG. 5A, according to one or more embodiments shown and described herein;
- (19) FIG. 5D schematically depicts a cross-sectional view of the pedal assembly of FIG. 5A taken from line 5-5 with a pedal arm in an undepressed state, according to one or more embodiments shown and described herein;
- (20) FIG. 5E schematically depicts a cross-sectional view of the pedal assembly of FIG. 5A taken from line 5-5 with a pedal arm in an partially depressed state, according to one or more embodiments shown and described herein;
- (21) FIG. 5F schematically depicts a cross-sectional view of the hanging pedal assembly of FIG. 5A taken from line 5-5 with a pedal arm in a fully depressed state, according to one or more embodiments shown and described herein;

- (22) FIG. 6A schematically depicts a right side view of a fourth example pedal assembly utilizing a pivot friction lever to generate resistance, according to one or more embodiments shown and described herein;
- (23) FIG. 6B schematically depicts a perspective view of the pedal assembly of FIG. 6A with a pedal arm in an undepressed state, according to one or more embodiments shown and described herein;
- (24) FIG. 6C schematically depicts a perspective view of the pedal assembly of FIG. 6A with a pedal arm in a partially depressed state, according to one or more embodiments shown and described herein;
- (25) FIG. 6D schematically depicts a perspective view of the pedal assembly of FIG. 6A with a pedal arm in a fully depressed state, according to one or more embodiments shown and described herein;
- (26) FIG. 7A schematically depicts a rear and right side perspective view of a fifth example pedal assembly utilizing a rotational pedal arm to generate resistance, according to one or more embodiments shown and described herein;
- (27) FIG. 7B schematically depicts a side view of the pedal assembly of FIG. 7A with the rotational pedal arm in an unengaged state, according to one or more embodiments shown and described herein;
- (28) FIG. 7C schematically depicts a side view of the pedal assembly of FIG. 7A with a pedal arm in a partially engaged state, according to one or more embodiments shown and described herein;
- and
- (29) FIG. 7D schematically depicts a side view of the pedal assembly of FIG. 7A with a pedal arm in a fully engaged state, according to one or more embodiments shown and described herein.

DETAILED DESCRIPTION

- (30) A brake pedal emulator (BPE) is a device that takes the place of a brake pedal and other hardware and is be used on an Electromechanical Braking System where there is no direct mechanical or hydraulic connection between the brake pedal and the calipers. The BPE inputs are force and travel distance from the driver's foot, reference voltage for all sensors, ground for all sensors, reaction loads at all fastening points. The BPE outputs are force feedback/resistance to driver's foot as a function of travel and speed, multiple pedal position sensor outputs as a function of travel, and error codes relating to the sensor outputs. Optional function is the conditioning of the output signals to provide the driver's intended braking input signal. The intention is that the BPE behaves to the driver as closely as possible as a conventional braking system in terms of pedal feel and deceleration performance.
- (31) The BPE may be located in a driver's footwell area. The BPE needs to meet the same mechanical loads as conventional pedal assemblies and must behave in a similar way as the conventional pedal. For example, the BPE needs to behave similar to conventional pedals when respect to applying loads, lateral loads, reverse loads vs. deflections and plastic deformation.
- (32) Conventional brake pedal assemblies include a pedal mounting bracket with a pivotally attached pedal arm/lever that has certain pedal force characteristics that need to be met during the apply stroke of the pedal. As such, the BPE needs to be configured to meet these same certain pedal force characteristics. Further, in some embodiments, the BPE may also include a downstop for the brake pedal stroke. Additionally, the BPE needs to be configured to withstand panic braking loads.
- (33) The BPE assemblies disclosed herein meet the following criteria: The BPE fails functional such that upon any failure, the driver is permitted to operate the braking system by applying the pedal and provide an appropriate sensor signal output. The BPE is configured to withstand foreseeable conditions and abuse a pedal will take. The BPE is scalable to automotive volume series production and be cost effective to manufacture and assemble.
- (34) Embodiments described herein are directed to a pedal emulator assemblies that include sensors adapted to sense a position and/or force of a pedal pad. The pedal emulator assemblies are

configured to simulate a braking fluid system, such as a hydraulic system, that is based on a speed system. That is, the faster a driver depresses onto the pedal pad of a pedal, the harder or more difficult the pedal is to depress, known herein as hepatic force.

(35) As a pedal effort (PE) is applied to the pedal, a pedal arm pivots to allow for the pedal to travel. The pedal emulator assemblies apply an opposite emulator force (EF) to provide the driver with a resistive force that changes according to the speed in which the PE is applied. Generally, the pedal emulator assembly outputs three distinct force vs travel sections that correspond to a specific range of travel such that the driver feels different resistance or haptics based on how fast the pedal is being depressed.

(36) Various embodiments of pedal emulator assemblies are described in detail herein. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

(37) As used herein, the term “communicatively coupled” means that coupled components are capable of exchanging data signals and/or electric signals with one another such as, for example, electrical signals via conductive medium or a non-conductive medium, though networks such as via Wi-Fi, Bluetooth, and the like, electromagnetic signals via air, optical signals via optical waveguides, and the like.

(38) As used herein, the term “longitudinal direction” refers to the forward-rearward direction of the assembly (i.e., in the $\pm$ -X-direction depicted in FIG. 3A). The term “lateral direction” refers to the cross-assembly direction (i.e., in the $\pm$ -Y-direction depicted in FIG. 3A), and is transverse to the longitudinal direction. The term “vertical direction” or “below” or “above” refer to the upward-downward direction of the assembly (i.e., in the $\pm$ -Z-direction depicted in FIG. 3A).

(39) Referring initially to FIG. 1, an example graphical representation of a desired force response curve for a pedal assembly is depicted, such as the various pedal assemblies depicted in FIGS. 3A-7D. As can be seen, as the pedal travels apply force that is required to generate haptic feedback increases. The force is non-linear and increases significantly near the end of the travel of the pedal. This type of force response is typically found in a mechanical pedal design where there is a linkage either mechanically or hydraulically coupled with the brake calipers. As a pedal effort (PE) is applied to the pedal, the pedal arm pivots to allow the pedal to travel. The emulator applies an opposite emulator force (EF) to provide the driver with a resistive force that changes according to the speed in which the PE is applied.

(40) For example, FIG. 1 depicts three distinct forces that correspond to a specific range of travel such that a driver of a vehicle feels varying resistance and/or haptics based on how fast a pedal pad of the pedal assembly is depressed. In the first section, illustrated by bracket **102**, the pedal arm is partially depressed by a first predetermined load or force, but minimal force is transferred from the pedal arm to other components of the emulator assemblies discussed herein.

(41) In the second section, as depicted by bracket **104**, upon a second predetermined amount of force applied on the pedal assembly to have an approximate, and without limitation, 30-60 millimeters of travel of a pedal arm, the slope needs to increase. As such, this slope is desirable when the various pedal assemblies described herein are in a plurality of semi-compressed states or partially compressed states. As illustrated, the upward slope of the curve depicted in FIG. 1 indicates that the greater distance that a pedal pad of the pedal assembly travels, the greater force acts on the pedal pad, such that a foot of the driver feels a desired haptic response such as a force feedback. The bracket **102** depicts a first force feedback or first return force and the bracket **104** depicts a second force feedback or second return force that is felt by the driver while applying the second predetermined amount of load onto the pedal pad. As such, the second force feedback or second return force is a greater force than the first force feedback or first return force that is felt by the driver while applying the second predetermined amount of load onto the pedal pad. Further, the second predetermined amount of load onto the pedal pad is a greater load than the first predetermined amount of load onto the pedal pad.

(42) In these embodiments, it should be appreciated that the amount of pedal travel, depicted without limitation, as the last 10 millimeters of travel, which may correspond to a third section as depicted by bracket **106**, may produce the greatest increase in force when a third predetermined force is applied to the pedal pad. As such, the bracket **106** depicts a third force feedback or third return force that is felt by the driver while applying the third predetermined amount of load onto the pedal pad. As such, the third force feedback or third return force is a greater force than the second force feedback or second return force that is felt by the driver while applying the third predetermined amount of load onto the pedal pad. Further, the third predetermined amount of load onto the pedal pad is a greater load than the second predetermined amount of load onto the pedal pad.

(43) Various pedal assemblies and their corresponding emulators will be described in additional detail herein with reference to FIGS. **3A-7D**.

(44) Referring now to FIG. **2**, an example graphical representation of a desired mechanical hysteresis force in terms of a force as a function of a pedal position is depicted. As can be seen in the figure, the curve has a similar shape to the apply force denoted in FIG. **1**. For example, the return force depicted in FIG. **2** may have an upward slope that corresponds to the upward slope of the target force per pedal travel distance depicted in FIG. **1**. It should be appreciated that, in these embodiments, it is desirable to have such force responses in a drive by wire braking system without a mechanical linkage. Such a system will include a brake pedal emulator that is substitute for a mechanically linked brake pedal. As noted hereinabove, various embodiments of brake pedal emulators will be described in additional detail herein with reference to FIGS. **3A-7D**.

(45) Referring now to FIGS. **3A-3F**, there is shown a schematic representation of a first embodiment of a pedal emulator assembly **300** that utilizes a sliding friction force to generate resistance. In this embodiment, the pedal emulator assembly **300** is depicted as a floor mounted pedal assembly. This is non-limiting and the pedal assembly may be a hanging type, organ type, or other type of pedal assembly. The pedal emulator assembly **300** includes a pedal arm **302** and a housing **304**. The pedal arm **302** includes a pivot end **306** and an opposite pad end **308** in which a pedal pad **310** is attached thereto. The pedal arm **302** includes an exterior surface **312a** and an opposite interior surface **312b**. A cavity **313** of the pedal arm **302** is defined by a pair of opposing walls **315a**, **315b**, an exterior wall **315c**, and the interior surface **312b** of the pedal arm **302**. The interior surface **312b** is positioned to face a cavity **314** of the housing **304** and portions of the pair of opposing walls **315a**, **315b** are received within the cavity **314** of the housing **304**. In some embodiments, the interior surface **312b** of the pedal arm **302** may not be a uniform, linear, and/or smooth contour surface and instead, the contour of the interior surface **312b** may include ramped portions **317a**, arcuate portions **317b**, raised portions **317c**, notches **317d**, and/or the like. In other embodiments, the interior surface **312b** of the pedal arm **302** may be linear or uniform contour. The cavity **314** of the housing **304** may be defined by a pair of sidewalls **316a**, **316b**, a first end wall **316c**, and a second end wall **316d**. Further, the cavity includes an interior surface **319** formed by the pair of sidewalls **316a**, **316b**, a first end wall **316c**, and a second end wall **316d**.

(46) In the embodiments described herein, the housing **304**, including sidewalls **316**, **316b**, first end wall **316c**, and second end wall **316d**, may be a molded plastic. For example, the housing **304** may be formed with various materials such as acrylonitrile butadiene styrene (ABS), polyethylene (PE), polypropylene (PP), polycarbonate (PC), nylon, polycarbonate/acrylonitrile butadiene styrene, polyurethane, polymethyl methacrylate, high density polyethylene, low density polyethylene, polystyrene, PEEK, POM (Acetal/Delrin), polyethylene terephthalate, thermoplastic elastomer, polyetherimide, thermoplastic vulcanizate, polysulfone, combinations thereof, and/or the like. Additionally, additives may be added such as UV absorbers, flame-retardants, colorants, glass fibers, plasticizers and/or the like.

(47) In other embodiments, the housing **304** and pedal arm **302** may be formed from injection molding or other additive manufacturing techniques. For example, as provided herein, additive

manufacturing techniques refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present invention may use layer-additive processes, layer-subtractive processes, or hybrid processes.

(48) Furthermore, it should be appreciated that the housing **304** may be floor mounted. That is, in some embodiments, the housing **304** may be coupled or mounted to be positioned within or extending from a floor surface of a vehicle. As such, the cavity **314** of the housing **304** accommodates the pedal arm **302** at full depression to allow the pedal pad **310** to fully travel.

(49) Referring still to FIGS. 3A-3F, the pedal emulator assembly **300** may further include a lever arm **318** that is positioned within the cavity **314** of the housing **304** and is pivotally coupled between the pair of sidewalls **316a**, **316b**. The lever arm **318** includes a first end **322a** and an opposite second end **322b**. Further, the lever arm **318** includes an upper frame **330** and a lower frame **332** that connect at an apex **328** at the first end **322a** and a hub portion **336** at the second end **322b**. A plurality of support members **334** extend between the upper frame **330** and the lower frame **332** to ensure the integrity of the lever arm **318** as the pedal arm **302** is depressed.

(50) As further depicted in FIGS. 3A-3F, the hub portion **336** positioned at the second end **322b** of the lever arm **318** is pivotally connected to the pair of sidewalls **316a**, **316b** of the housing **304** via an attachment portion **321** extending between the pair of sidewalls **316a**, **316b**. The attachment portion **321** may include a pair of receiving slots **323** configured to receive a circumferential wall **325** or continuous circular wall of the lever arm **318** extending from the second end **322b** in the lateral direction (i.e., in the $\pm Y$ direction). For example, the hub portion **336** may be coupled to the pair of sidewalls **316a**, **316b** using a press fit, a through pin, and the like. In other embodiments, a protrusion may extend from each of the pair of sidewalls **316a**, **316b** to be received by the hub portion **336**, such that the hub portion **336** may pivot relative the pair of sidewalls **316a**, **316b**. However, it should be understood that the hub portion **336** may be coupled to the pair of sidewalls **316a**, **316b** via any suitable coupling that allows the hub portion **336** to pivot. By pivotally coupling the hub portion **336** between the pair of sidewalls **316a**, **316b**, the lever arm **318** may rotate about the hub portion **336** when the interior surface **312b** of the pedal arm **302** contacts the upper frame **330** of the lever arm **318**, as will be described in additional detail herein.

(51) As further depicted in FIGS. 3A-3F, the upper frame **330** includes a surface **331** that has nonlinear profile extending between the first end **322a** and the second end **322b** of the lever arm **318** and through an apex **340** of the upper frame **330**. For example, as depicted best in FIGS. 3D-3F, a first portion **327** of the upper frame **330** (e.g., the portion extending between the first end **322a** and the apex **340**) may have a first slope, while a second portion **329** of the upper frame **330** (e.g., the portion extending between the apex **340** and the second end **322b**) may have a second slope different from the first slope, which may provide the surface **331** of the upper frame **330** with the nonlinear profile. In these embodiments, the first slope of the first portion **327** may be greater (e.g., steeper) than the second slope of the second portion **329**, which may aid in ensuring that appropriate forces are provided as the pedal is depressed. Operation of the pedal emulator assembly **300** will be described in additional detail herein with reference to FIGS. 3D-3F.

(52) In the embodiments described herein, the lever arm **318** may further include a spring receiving portion **320** positioned between the first end **322a** and the second end **322b** in the lower frame **332**.

(53) A spring carrier **343** that includes a compressible member, such as at least one spring **326**, may

extend between the spring receiving portion **320** of the lever arm **318** and a spring retaining portion **324** formed in a lower arm **341**. The lower arm **341** may be fixedly coupled to the floor surface of the cavity, and/or to the interior surface **319**, and the like.

(54) The lower arm **341** further includes an upper portion **365a** and an opposite lower portion **365b**. The lower portion **365b** is coupled to the housing **304** via a lower arm protrusion **366** that extends from the lower portion **365b** of the lower arm **341**. In these embodiments, the lower arm protrusion **366** of the lower arm **341** is received in a recess **370** of the housing **304** formed in the second end wall **316d**. The upper portion includes an engagement surface **367** that is configured to receive a portion of the apex **328** of the joining of the upper frame **330** and the lower frame **332** of the lever arm **318** based on an amount of travel of the pedal arm **302**. For example, as best illustrated in FIG. **3F**, the apex **328** is illustrated as abuts, engaging, or otherwise in communication with the engagement surface **367** of the lower arm **341** when the pedal arm **302** is in the fully compressed position adding to the pedal effect felt by the driver.

(55) The lower arm **341** may further include a concave portion **361** configured to receive a pivot portion **363** that is rotatably or movably coupled to the **306** of the pedal arm **302** such that the pedal arm pivots or rotates or otherwise moves the pivot portion **363** within the concave portion **361** of the lower arm **341**.

(56) As such portion of the spring carrier **343** may be received within, the cavity **313** of the pedal arm **302** and a portion may be received within the cavity **314** of the housing **304**. In some embodiments, the spring carrier **343** may be coupled to spring receiving portion **320** of the lever arm **318** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **343** may be coupled to the spring receiving portion **320** of the lever arm **318** via a press fit configuration, a tension fit, and the like. Further, in some embodiments, the spring carrier **343** may be coupled to the spring retaining portion **324** of the lower arm **341** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **343** may be coupled to spring retaining portion **324** of the lower arm **341** via a press fit configuration, a tension fit, and the like.

(57) The spring carrier **343** further includes a female spring guide **345** and a male spring guide **347** that includes an elongated member **355** configured to extend and engage with the female spring guide **345** and compress into a void or cavity in the male spring guide **347** based on a tension or force applied to the spring carrier **343**. It should be understood that this is non-limiting and the elongated member **355** may be configured to extend and engage with the male spring guide **347** and compress into a void or cavity in the female spring guide **345** based on a tension or force applied to the spring carrier **343**.

(58) The female spring guide **345** and the male spring guide **347** are coupled to one another within an inner diameter of the at least one spring **326** (e.g., via the elongated member **355**) such that the at least one spring **326** extends between and is in contact with a spring receiving surface **349a** of the female spring guide **345** and a spring receiving surface **349b** of the male spring guide **347**. Such an arrangement retains the at least one spring **326** while permitting for the at least one spring **326** to expand and compress as a function of the amount of travel of the pedal arm **302**, as discussed in greater detail herein.

(59) A protrusion **351** extends from the male spring guide **347** to act as a coupling point to pivotally couple the male spring guide **347** to a corresponding slots **353** in the spring retaining portion **324** of the lower arm **341** in a press fit or tension fit connection. In other embodiments, the protrusion **351** of the male spring guide **347** may be coupled to the corresponding slots **353** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like.

(60) The tension caused by the at least one spring **326** arranged between the male spring guide **347** and the female spring guide **345** that are configured to move to compress the at least one spring **326** may cause the female spring guide **345** to remain in contact with or seated within the spring receiving portion **320** of the lever arm **318** and cause the male spring guide **347** to remain in

contact with, or seated within, the spring retaining portion **324** formed in the lower arm **341**. As such, regardless of the amount of travel of the pedal arm **302**, there is a tension caused by the male spring guide **347** and the female spring guide **345** maintaining a position of contact of the spring carrier **343** extending between the lever arm **318** and the lower arm **341**.

(61) The at least one spring **326** biases the lever arm **318** into the interior surface **312b** (e.g., via the first end **322a**, as has been described herein) with a force sufficient to maintain the pedal pad **310** in an initial or undepressed position, such that the pedal pad **310** is available for the driver, as best illustrated in FIG. 3D.

(62) Although the pedal emulator assembly **300** depicted in FIGS. 3A-3F is illustrated as including a single spring as the at least one spring **326**, it should be appreciated that, in some embodiments, multiple springs may extend between the spring receiving portion **320** and the spring retaining portion **324** formed in the lower arm **341**. The multiple springs may be coaxially aligned such that the at least one spring **326** has an outer diameter smaller or less than an inner diameter of the at least one spring **326** to be positioned within the inner diameter of the at least one spring **326**, and so on. For example, in some embodiments, multiple springs may be utilized to ensure that the pedal emulator assembly **300** remains functional in the event at least one of the springs fails (e.g., breaks, disengages the spring retaining portion and/or spring retention portion, and the like). In these embodiments, the at least one spring **326** may be a coil spring. In other embodiments, the at least one spring **326** may be a torsion spring, a tension spring, a conical spring, and/or the like. The at least one spring **326** may be formed with a steel material, such as stainless steel, wire, carbon steel, alloy steel, elgiloy, Monel®, copper, nickel, and/or the like.

(63) Operation of the pedal emulator assembly **300** will now be discussed in detail with reference to FIGS. 3D-3F. For example, as depicted in FIG. 3D, the pedal emulator assembly **300** is in the initial or undepressed position, and/or subject to a first predetermined load **L1**. In this position, the pedal pad **310** is extended away from the housing **304** in the longitudinal direction (e.g., in the $+/-X$ direction) such that the interior surface **312b** of the pedal arm **302** is only in contact with the first end **322a** of the lever arm **318**. In these embodiments, the at least one spring **326** may be in a mostly unbiased (e.g., fully extended) position, such that the at least one spring **326** is only applying a minimum resistance to maintain the positioning of the pedal arm **302**. As such, the first predetermined load **L1** may be enough to force to compress the at least one spring **326** a predetermined distance and the pedal arm may slidably contact or slidably engage with a portion of the non-linear profile of the surface **331** of the upper frame **330** of the lever arm **318** to achieve the desirable curve illustrated by bracket **102** in FIG. 1.

(64) Referring now to FIG. 3E, when a second predetermined load **L2** is applied to the pedal pad **310**, in the direction indicated by arrow **A1**, the pedal arm **302** moves or pivots about the pivot end **306**, indicated by arrow **A2**, to allow the pad end **308** to move into the housing **304** based on the amount of force exerted on the pedal pad **310** and the current position of the pedal arm **302**. For example, as depicted in FIG. 3E, the pad end **308** may move partially into the housing **304**, such that the at least one spring **326** is partially compressed and generates a return force or a force feedback on the pedal pad **310** that corresponds to the amount of force applied to the pedal pad **310** by the driver.

(65) Simultaneously, the interior surface **312b** of the pedal arm **302** may abut or ride against the surface **331** of the upper frame **330** in addition the compression of the at least one spring **326**. In these embodiments, the amount of force applied to the pedal pad **310** (e.g., via the driver), may correspond to the pedal effort and/or resistance felt by the driver during depression of the pedal pad **310**. More particularly, the mechanical advantage of the pedal arm **302** against the at least one spring **326** may determine the pedal effort and/or resistance felt by the driver during depression of the pedal pad **310**. In these embodiments, the mechanical advantage between the pedal arm **302** and the at least one spring **326** refers to the ratio of the force exerted by the at least one spring **326** relative to the force applied to the pedal arm **302** via contact between the driver and the pedal pad

310.

(66) Referring still to FIG. 3E, a greater or larger portion of the non-linear profile of the surface **331** of the upper frame **330** of the lever arm **318** may further ride against or slidably contact or slidably engage with interior surface **312b** of the pedal arm **302** producing a greater pedal effect experienced by the driver, as illustrated by bracket **104** in FIG. 1. Further, the at least one spring **326** may still be compressing providing an additional pedal effect. In these embodiments, the effective force exerted by the at least one spring **326** may depend on the contour of the surface **331** of the upper frame **330** relative to the contour of the interior surface **312b** of the pedal arm **302** contacting the upper frame **330**.

(67) For example, in the embodiments depicted in FIGS. 3D-3F, the contour of the interior surface **312b** of the pedal arm **302** may be similar to the contour of the first portion of the upper frame **330** (e.g., the portion between the first end **322a** and the apex **340**). Accordingly, when the pedal arm **302** is partially depressed, as best depicted in FIG. 3E, the force exerted on the driver's foot may be less, as the similar contours of the interior surface **312b** of the pedal arm **302** and the first portion of the upper frame **330** may act to balance the force exerted by the at least one spring **326**. In these embodiments, the force exerted on the foot of a driver may simulate the initial give or ease of a traditional brake pedal, which is depicted in the bracket **104** of FIG. 1.

(68) As the pedal arm **302** is further depressed, the interior surface **312b** of the pedal arm **302** may engage different portions of the upper frame **330**. As the contour of the upper frame **330** changes, the upper frame **330** may provide increased mechanical advantage against the at least one spring **326**, thereby simulating a progressive increase in resistance or pedal effort felt by the driver.

(69) The mechanical advantage against the at least one spring **326** may be increased as the contour of the upper frame **330** departs from the contour of the interior surface **312b** of the pedal arm **302**. For example, as depicted in FIGS. 3D-3F, the second portion **329** of the upper frame **330** (e.g., the portion between the apex **340** and the second end **322b**) may greatly differ in slope from the interior surface **312b** of the pedal arm **302**. Accordingly, as the interior surface **312b** of the pedal arm **302** contacts the second portion **329** of the upper frame **330**, the force provided by the at least one spring **326** acts at an angle, thereby increasing the mechanical advantage or pedal effort against the pedal arm **302**.

(70) For example, as the pedal arm **302** moves from the partially depressed position (as depicted in FIG. 3E) to the fully depressed position (as depicted in FIG. 3F), the interior surface **312b** of the pedal arm **302** may transition from engaging the first portion **327** of the upper frame **330** to the second portion **329** of the upper frame **330**. As the interior surface **312b** of the pedal arm **302** contacts the second portion **329** of the upper frame **330**, the mechanical advantage and/or pedal effort against the at least one spring **326** may be maximized, thereby maximizing the force exerted back onto the pedal arm **302** (and thus, the foot of the driver).

(71) Accordingly, it should be appreciated that the various contours of the interior surface **312b**, the surface **331** of the upper frame **330** of the lever arm **318** and the at least one spring **326** provide the pedal effect/resistance felt at the pedal pad **310** by the driver. The force applied by the at least one spring **326** and the various contours of the surface **331** of the interior surface **312b** applied against the non-linear surface of the upper frame **330** of the lever arm **318** apply different forces at the corresponding travel positions of the pedal arm **302** with the surface **331** having the non-linear contour applying a desirable force to the pedal arm **302**. The combination of the at least one spring **326** and the various contours of the interior surface **312b** applied against the surface **331** of the upper frame **330** of the lever arm **318** result in a composite of forces to provide a desired force curve.

(72) Referring again to FIGS. 3A-3F, in the embodiments described herein, the pedal emulator assembly **300** may further include a sensor assembly **350**. In these embodiments, the sensor assembly **350** may include a printed circuit board including at least one Hall Effect chip and a magnet positioned anywhere within the housing **304** that moves to detect movement of the pedal

arm **302** and/or the at least one spring **326** via Hall Effect sensing techniques. Accordingly, the sensor assembly **350** may be configured to track the position of the pedal arm **302** and the forces applied to the foot of the driver. In other embodiments, inductive sensing techniques may be applied to sense a coupler that moves to detect movement of the pedal arm **302** and/or the at least one spring **326**.

(73) Turning now to FIGS. **4A-4E**, a second example embodiment of a pedal emulator assembly **400** that utilizes friction to generate resistance is schematically depicted. In these embodiments, the pedal emulator assembly **400** includes a pedal arm **402** and a housing **404**. The pedal arm **402** includes a pivot end **406** and an opposite pad end **408** in which a pedal pad **410** is attached thereto. The pedal arm **402** includes an exterior surface **412a** and an opposite interior surface **412b**. A protrusion **413** extends from the interior surface **412b** of the pedal arm **402**. In some embodiments, the protrusion **413** may be shaped to have a contact surface **415** that is rounded. In other embodiments, the protrusion **413** may be any shape including, without limitation, octagonal, hexagonal, triangular, cone, square, and/or the like. A portion of the pedal arm **402** may be received within a cavity **414** of the housing **404**.

(74) The cavity **414** of the housing **404** is defined by a pair of sidewalls **416a**, **416b**, an end wall **416c**, and a top wall **416d**. The pivot end **406** of the pedal arm **402** is pivotally connected to the pair of sidewalls **416a**, **416b** via an elongated member **417**, such that the pedal arm **402** may rotate, in the direction indicated by arrow **A4**, about the pivot end **406**. The housing **404** is configured to be mounted to an instrument panel, a bracket, and/or the like, such that the housing **404** is supported above a floor of a vehicle. Within the cavity **414** of the housing **404** is a lever arm **418** pivotally coupled between the pair of sidewalls **416a**, **416b**. The lever arm **418** includes a first end **420a** and an opposite second end **420b**. Further, the lever arm **418** includes an outer surface **419a** and an opposite inner surface **419b** that are spaced apart by a plurality of honeycomb members **419c**. Further, the outer surface **419a** includes an engagement surface **422** positioned to extend along the outer surface **419a** from the first end **420a**. The engagement surface **422** may be configured to abut and or receive the contact surface **415** of the protrusion such that the protrusion **413** slidably engages with the engagement surface **422**. Both the outer surface **419a** and the engagement surface **422** are non-linear and may have portions that are accurate and/or curvilinear.

(75) Further, the lever arm **418** includes a projection **423** extending from inner surface **419b** that is configured to be received within a recess **427** of an interior surface **429** of the end wall **416c** within the cavity **414**. As such, the interaction between the projection **423** and the recess **427** permit the lever arm **418** to pivot or otherwise move within the cavity **414** a predetermined amount based on a load applied to the pedal pad **410**.

(76) A spring carrier **426** includes a pedal end **428a** and an opposite lever arm end **428b**. The spring carrier **426** extends between the pivot end **406** of the pedal arm **402** and the first end **420a** of the lever arm **418**. The pedal end **428a** is coupled to a hub receiving portion **430** that is coupled to the pivot end **406** of the pedal arm **402** and the lever arm end **428b** is coupled to the first end **420a** of the lever arm **418**. The second end **420b** of the lever arm **418** is adapted to receive at least one spring **436** that extends between the lever arm **418** and the hub receiving portion **430**.

(77) In some embodiments, the spring carrier **426** may be coupled to a spring receiving portion **421** of the lever arm **418** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **426** may be coupled to the spring receiving portion **421** of the lever arm **418** via a press fit configuration, a tension fit, and the like. Further, in some embodiments, the spring carrier **426** may be coupled to a spring carrier retaining portion **438** of the pedal arm **402** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **426** may be coupled to the spring carrier retaining portion **438** of the pedal arm **402** via a press fit configuration, a tension fit, and the like.

(78) The spring carrier **426** further includes a female spring guide **445** and a male spring guide **447**

that includes an elongated member **455** configured to extend and engage with the female spring guide **445** and compress into a void or cavity in the male spring guide **447** based on a tension or force applied to the spring carrier **426**. It should be understood that this is non-limiting and the elongated member **455** may be configured to extend and engage with the male spring guide **447** and compress into a void or cavity in the female spring guide **445** based on a tension or force applied to the spring carrier **426**. The male spring guide **447** includes a first end **426a** and an opposite second end **426b**, the first end **426a** configured to engage with the pedal arm at the pivot end **406** and the second end **426** is positioned at the female spring guide **445**.

(79) The female spring guide **445** and the male spring guide **447** are coupled to one another within an inner diameter of the at least one spring **436** (e.g., via the elongated member **455**) such that the at least one spring **436** extends between and is in contact with a spring receiving surface **449a** of the female spring guide **445** and a spring receiving surface **449b** of the male spring guide **447**. Such an arrangement retains the at least one spring **436** while permitting for the at least one spring **436** to expand and compress as a function of the amount of travel of the pedal arm **402**, as discussed in greater detail herein.

(80) The tension caused by the at least one spring **436** arranged between the male spring guide **447** and the female spring guide **445** that are configured to move to compress the at least one spring **436** may cause the female spring guide **445** to remain in contact with or seated within the spring receiving portion **421** of the lever arm **418** and cause the male spring guide **447** to remain in contact with, or seated within, the spring carrier retaining portion **438** formed in the pedal arm **402**. As such, regardless of the amount of travel of the pedal arm **402**, there is a tension caused by the male spring guide **447** and the female spring guide **445** maintaining a position of contact of the spring carrier **426** extending between the lever arm **418** and the pedal arm **402**.

(81) Referring still to FIGS. **4A-4E**, the pedal emulator assembly **400** further includes an elastomer member **432** positioned within a recess **434** of the housing **404**. The recess **434** is positioned below the cavity **414** such that the elastomer member **432** is positioned to align with the interior surface **412b** of the pedal arm **402** at the pad end **408** such that when the pedal arm **402** is moved or rotated a predetermined distance, the interior surface **412b** of the pedal arm **402** at the pad end **408** makes contact to compress the elastomer member **432**.

(82) In the embodiments described herein, the housing **404** and pedal arm **402**, may be a molded plastic. For example, the housing **404** and pedal arm **402** may be formed with various materials such as acrylonitrile butadiene styrene (ABS), polyethylene (PE), polypropylene (PP), polycarbonate (PC), nylon, polycarbonate/acrylonitrile butadiene styrene, polyurethane, polymethyl methacrylate, high density polyethylene, low density polyethylene, polystyrene, PEEK, POM (Acetal/Delrin), polyethylene terephthalate, thermoplastic elastomer, polyetherimide, thermoplastic vulcanizate, polysulfone, combinations thereof, and/or the like. Additionally, additives may be added such as UV absorbers, flame-retardants, colorants, glass fibers, plasticizers and/or the like.

(83) In other embodiments, the housing **404** and pedal arm **402** may be formed from injection molding or other additive manufacturing techniques. For example, as provided herein, additive manufacturing techniques refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present invention may use layer-

additive processes, layer-subtractive processes, or hybrid processes.

(84) Operation of the pedal emulator assembly **400** will now be described in detail with reference to FIGS. **4D-4F**. Initially, FIG. **4D** depicts the pedal emulator assembly **400** in an uncompressed (e.g., initial or default) position, and/or subject to a first predetermined load **L1**. In these embodiments, the lever arm **418** is also positioned in an initial position, such that the interior surface **412b** of the pedal arm applies only a minimum force on the lever arm **418**. As such, the first predetermined load **L1** may be enough force to compress the pedal arm **402** a predetermined distance such that the pedal arm **402** slidably contacts or slidably engages the engagement surface **422** of the lever arm to achieve the desirable curve illustrated by bracket **102** in FIG. **1**.

Accordingly, it should be further appreciated that, in when the first predetermined load **L1** is applied to the pedal arm, as depicted in FIG. **4D**, the at least one spring **436** and the elastomer member **432** positioned within the recess **434** of the housing **404** is similarly uncompressed.

(85) When a second predetermined load **L2** that is greater than the first predetermined load **L1** is applied to the pedal pad **410**, in the direction indicated by arrow **A3**, the pedal arm **402** moves or pivots about the pivot end **406**, indicated by arrow **A4**, based on the amount of force applied to the pedal pad **410**. In these embodiments, the coupling of the pivot end **406** to the pair of sidewalls **416a**, **416b** allows the pedal arm **402** to move in an arcuate trajectory. Simultaneously, the motion of the pedal arm **402** is transferred via the spring carrier **426** to compress the at least one spring **436**. In these embodiments, the second predetermined load **L2** applied to the pedal (e.g., in the direction indicated by arrow **A3**) may cause the pedal pad **410** to move to a partially compressed position that corresponds to the second bracket **104** depicted in FIG. **1**.

(86) Referring still to FIG. **4E**, and as described herein, the outer surface **419a** of the lever arm **418** may be configured to interact with the interior surface **412b** of the pedal arm **402**. Because each of the outer surface **419a** of the lever arm **418** and the interior surface **412b** of the pedal arm **402** are non-linear surfaces, different portions of the outer surface **419a** of the lever arm **418** contact the interior surface **412b** of the pedal arm **402** as the pedal arm **402** is depressed, with each of the portion of the interior surface **412b** of the pedal arm providing a variable amount of frictional force on the engagement surface **422** of the lever arm **418**.

(87) In the embodiments described herein, it should be appreciated that the variability in the frictional force acting on the outer surface **419a** of the lever arm **418** may be used to produce a desired force curve. For example, as the pedal arm **402** is depressed, the resistance the pedal arm **402** offers the driver is not consistent and/or linear; instead, the resistance of the pedal arm **402** varies to emulate the feeling of a traditional vehicle pedal. In these embodiments, it may be possible to manipulate the surfaces of the outer surface **419a** of the lever arm **418** and the engagement surface **422** in order to obtain the desired force curve depicted in FIGS. **1** and **2**, as will be described herein.

(88) For example, as the pedal arm **402** is depressed, the pivoting motion of the pedal arm **402** is translated to cause the lever arm **418** to pivot. For example, as the pedal arm **402** rotates, the interior surface **412b** of the pedal arm **402** slides against the engagement surface **422** of the pedal arm **402**, such that the lever arm **418** pivots. Accordingly, it should be appreciated that the amount of contact between the outer surface **419a** of the lever arm **418** and the interior surface **412b** of the pedal arm may determine the frictional force that exists between the two components, and in turn, the resistance offered by the pedal arm **402**.

(89) In order to achieve the desired force curve depicted in FIGS. **1** and **2**, the interior surface **412b** of the pedal arm and the outer surface **419a** of the lever arm **418** may be configured such that the two surfaces increasingly contact one another as the pedal arm **402** is depressed. For example, a portion of the outer surface **419a** of the lever arm **418** and/or the interior surface **412b** of the pedal arm may be textured, include a plurality of bumps, or otherwise include non-linear portions (e.g., curved) such that contact between the interior surface **412b** of the pedal arm **402** and the outer surface **419a** of the lever arm **418** increases as the pedal arm **402** is depressed. Accordingly, the

increase in friction between the interior surface **412b** of the pedal arm **402** and the outer surface **419a** of the lever arm **418** as the pedal arm **402** is depressed may similarly increase the resistance offered by the pedal arm **402**, thereby generating the desired force curves depicted in FIGS. **1** and **2**.

(90) Referring still to FIG. **4E**, when the second predetermined load **L2** is applied to the pedal arm **402**, the second end **420b** of the lever arm **418** (e.g., the end connected to the spring mechanism) may compress the spring to further add to the resistance offered by the pedal arm **402**. As should be appreciated, as the pedal arm **402** is depressed, and the lever arm **418** continues to pivot, the spring may become increasingly compressed, thereby increasing the resistance offered by the pedal arm **402**. In these embodiments, variations in the type of spring and the compression characteristics of said spring may further impact or influence the resistance offered by the pedal arm **402** as the pedal arm **402** is depressed.

(91) Turning now to FIG. **4F**, the pedal emulator assembly **400** is depicted in a fully compressed position, in which a third predetermined load **L3** is applied to the pedal arm **402**. In these embodiments, as the third predetermined load **L3** is applied to the pedal pad **410**, the interior surface **412b** of the pedal arm **402** makes contact with and compresses the elastomer member **432**. In these embodiments, as the pedal arm **402** contacts the elastomer member **432**, the elastomer member **432** may resist compression, thereby offering additional resistance to the pedal arm **402**. In particular, the additional resistance offered by the elastomer member **432** may correspond to the third section depicted by bracket **106** of the desired force curve shown in FIG. **1**.

(92) Referring again to FIGS. **4A-4F**, in the embodiments described herein, the pedal emulator assembly **400** may further include a sensor assembly **450**. In these embodiments, the sensor assembly **450** may include a printed circuit board including at least one Hall Effect chip and a magnet positioned anywhere within the housing **404** that moves to detect movement of the pedal arm **402** and/or the lever arm **418**, via Hall Effect sensing techniques. In other embodiments, inductive sensing techniques may be applied to sense a coupler that moves to detect movement of the pedal arm **302** and/or the at least one spring **326**. Accordingly, the sensor assembly **450** may be configured to track the position of the pedal arm **402** and the forces applied to the foot of the driver.

(93) In view of the foregoing, it should be appreciated that, the spring, the non-linear contour of the outer surface **419a** and the engagement surface **422** and the elastomer member **432** provide the pedal effort/resistance felt by the driver during the depression of the pedal pad **310**. That is, the various contours of the outer surface **419a** and the engagement surface **422**, the at least one spring, and the elastomer member **432** provide the pedal effect/resistance felt at the pedal pad **410** by the driver and apply different forces at the corresponding travel positions with the elastomer member **432** applying a significant force to the pedal arm **402**. The combination of the contours of the outer surface **419a** and the engagement surface **422**, the at least one spring, and the elastomer member **432** result in a composite of forces to provide a desired force curve.

(94) Turning now to FIGS. **5A-5F**, a third example embodiment of a pedal emulator assembly **500** that utilizes a slot to generate friction resistance is schematically depicted. The pedal emulator assembly **500** includes a pedal arm **502** and a housing **504**. The pedal arm **502** includes a pivot end **506** and an opposite pad end **508** in which a pedal pad **510** is attached thereto.

(95) The pedal arm **502** includes an exterior surface **512a** and an opposite interior surface **512b**. The cavity **513** of the pedal arm **502** is defined by a pair of opposing walls **515a**, **515b**, an exterior wall **515c**, and the interior surface **512b** of the pedal arm **502**. The interior surface **512b** is positioned to face a cavity **514** of the housing and portions of the pair of opposing walls **515a**, **515b** are received within the cavity **514** of the housing **504**. The cavity **514** of the housing **504** is defined by a pair of sidewalls **516a**, **516b**, a first end wall **516c**, and a second end wall **516d**. Further, the cavity includes an interior surface **519** formed by the pair of sidewalls **516a**, **516b**, the first end wall **516c**, and the second end wall **516d**.

(96) In some embodiments, the housing **504** may be configured to be mounted to a floor of a

vehicle. In other embodiments, the housing **504** may be configured to be mounted to any component of the vehicle within a passenger compartment such as an instrument panel, a firewall, and the like.

(97) In the embodiments described herein, the housing **504** and pedal arm **502** may be a molded plastic. For example, the housing **504** and pedal arm **502** may be formed with various materials such as acrylonitrile butadiene styrene (ABS), polyethylene (PE), polypropylene (PP), polycarbonate (PC), nylon, polycarbonate/acrylonitrile butadiene styrene, polyurethane, polymethyl methacrylate, high density polyethylene, low density polyethylene, polystyrene, PEEK, POM (Acetal/Delrin), polyethylene terephthalate, thermoplastic elastomer, polyetherimide, thermoplastic vulcanizate, polysulfone, combinations thereof, and/or the like. Additionally, additives may be added such as UV absorbers, flame-retardants, colorants, glass fibers, plasticizers and/or the like.

(98) In other embodiments, the housing **504** and pedal arm **502** may be formed from injection molding or other additive manufacturing techniques. For example, as provided herein, additive manufacturing techniques refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present invention may use layer-additive processes, layer-subtractive processes, or hybrid processes.

(99) Referring still to FIGS. 5A-5F, the pedal emulator assembly **500** may further include a pair of slots **518** that extend through the pair of opposing walls **515a**, **515b**, respectively, between the pivot end **506** and the pad end **508** and below the interior surface **512b** of the pedal arm **502** in the vertical direction (i.e., in the $\pm Z$ direction). Each of the pair of slots **518** extend in the longitudinal direction (i.e., in the $\pm X$ direction). In these embodiments, a link lever arm **520** that has a support end **522** and a coupled end **524** is positioned to within the cavity **513**, **514** and to be movably coupled to the pair of slots **518**. The coupled end **524** may be pivotally or movably coupled to a lower arm **526**. The support end **522** includes a pair of ears **528** extending from a body **530**. Each of the pair of ears **528** are spaced apart and/or dimensioned to be aligned with the pair of slots **518** in the lateral direction (i.e., in the $\pm Y$ direction). Each ear of the pair of ears **528** include an aperture **532** extending there through. The apertures **532** receive an elongated member **533** with a circumferential outer surface **535** that has a diameter that is smaller than the apertures **532**. The circumferential outer surface **535** may be a smooth contour and configured to slidably engage with the pair of slots **518** and with an upper lever arm **534**, as discussed in greater detail herein. The elongated member **533** that has a pair of terminating ends **537** with a diameter that is larger than each of the pair of slots **518**. In some embodiments, the elongated member **533** may be a through pin or other elongated type member to slidably couple the support end **522** of the link lever arm **520** to the pedal arm **502**. The coupled end **524** may be pivotally coupled to the lower arm **526**, such that the link lever arm **520** may pivot or otherwise move relative the lower arm **526** dependent on an amount of load applied to the pedal arm **502** as will be described in additional detail herein with reference to FIGS. 5D-5F.

(100) As further depicted in FIGS. 5A-5F, the upper lever arm **534** is pivotally coupled between the pair of sidewalls **516a**, **516b**. The upper lever arm **534** includes a first end **536a** and an opposite second end **536b**. Further, the upper lever arm **534** includes an upper frame portion **538** and a lower frame portion **540** spaced apart and coupled by a plurality of honeycomb support members **544**.

The upper frame portion **538** and the lower frame portion **540** may connect or abut at an apex **542** that may be positioned at and/or between the pair of ears **528** of the support end **522** of the link lever arm **520** when a first predetermined load **L1** is applied to the pedal pad **510**. The second end **536b** is pivotally connected to the pair of sidewalls **516a**, **516b**. In some embodiments, a through pin **543** may be used to pivotally couple or otherwise movably couple the second end **536b** to the pair of sidewalls **516a**, **516b**. In other embodiments, a protrusion may extend from each of the pair of sidewalls **516a**, **516b** to be received by the second end **546b**. The upper lever arm **534** further includes a spring receiving portion **546** positioned between the first end **536a** and the second end **536b** and extending within the lower frame portion **540**.

(101) Referring still to FIGS. 5A-5F, the upper frame portion **538** includes an engagement surface **549** and an upper surface **550**. The engagement surface **549** may have a nonlinear profile, or counter, that is adapted to receive, abut, or otherwise slidably engages with the circumferential outer surface **535** of the elongated member **533** such that the circumferential outer surface **535** rides along the engagement surface **549** based on the amount of the load applied to the pedal pad **510**, as will be described in additional detail herein.

(102) The lower arm **526** includes a first end **546a** that extends from and is pivotally coupled to the pivot end **506** of the pedal arm **502** and an opposite second end **546b**. The lower arm **526** includes a spring retaining portion **547** positioned at or adjacent to the second end **546b**. A spring carrier **570** that includes a compressible member, such as at least one spring **548**, extends between the spring retaining portion **547** of the lower arm **526** and the spring receiving portion **546** of the upper lever arm **534** to bias the upper lever arm **534** in a direction opposite of a pedal depression force (e.g., in the opposite direction depicted by the arrow **A5** in FIGS. 5A and 5B). In these embodiments, the lower arm **526** may be fixedly coupled to the floor surface of the cavity **513**, and/or to the interior surface **519**, and the like.

(103) The lower arm **526** further includes an upper portion **526a** and an opposite lower portion **526b**. The lower portion **526b** is coupled to the housing **504** via a lower arm protrusion **566** that extends from the lower portion **526b** of the lower arm **526**. In these embodiments, the lower arm protrusion **566** of the lower arm **526** is received in a recess **571** of the housing **504** formed in the second end wall **516d**. The upper portion includes an engagement surface **567** that is configured to receive a portion of the link lever arm **520**, such as the coupled end **524** of the link lever arm **520**, based on an amount of travel of the pedal arm **502**. For example, as best illustrated in FIG. 5F, the coupled end **524** of the link lever arm **520** is illustrated as pivoting about, or otherwise being in communication with the engagement surface **567** of the lower arm **526** when the pedal arm **502** is in the fully compressed position adding to the pedal effect felt by the driver.

(104) The lower arm **526** may further include a concave portion **561** configured to receive a pivot portion **563** that is rotatably or movably coupled to the pivot end **506** of the pedal arm **502** such that the pedal arm pivots or rotates or otherwise moves the pivot portion **563** within the concave portion **561** of the lower arm **526**.

(105) As such portion of the spring carrier **570** may be received within the cavity **513** of the pedal arm **502** and a portion may be received within the cavity **514** of the housing **504**. In some embodiments, the spring carrier **570** may be coupled to spring receiving portion **546** of the link lever arm **520** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **570** may be coupled to the spring receiving portion **546** of the link lever arm **520** via a press fit configuration, a tension fit, and the like. Further, in some embodiments, the spring carrier **570** may be coupled to the spring retaining portion **547** of the lower arm **526** via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like. In other embodiments, the spring carrier **570** may be coupled to spring retaining portion **547** of the lower arm **526** via a press fit configuration, a tension fit, and the like.

(106) The spring carrier **570** further includes a female spring guide **575** and a male spring guide

577 that includes an elongated member 555 configured to extend and engage with the female spring guide 575 and compress into a void or cavity in the male spring guide 577 based on a tension or force applied to the spring carrier 570. It should be understood that this is non-limiting and the elongated member 555 may be configured to extend and engage with the male spring guide 577 and compress into a void or cavity in the female spring guide 575 based on a tension or force applied to the spring carrier 570.

(107) The female spring guide 575 and the male spring guide 577 are coupled to one another within an inner diameter of the at least one spring 548 (e.g., via the elongated member 555) such that the at least one spring 548 extends between and is in contact with a spring receiving surface 575a of the female spring guide 575 and a spring receiving surface 577b of the male spring guide 577. Such an arrangement retains the at least one spring 548 while permitting for the at least one spring 548 to expand and compress as a function of the amount of travel of the pedal arm 502, as discussed in greater detail herein.

(108) A protrusion 551 extends from the male spring guide 577 to act as a coupling point to pivotally couple the male spring guide 577 to corresponding slots 553 in the spring retaining portion 547 of the lower arm 526 in a press fit or tension fit connection. In other embodiments, the protrusion 551 of the male spring guide 577 may be coupled to the corresponding slots 553 via a fastener such as a nut and bolt, screw, rivet, hook and loop, adhesive, weld, and/or the like.

(109) The tension caused by the at least one spring 548 arranged between the male spring guide 577 and the female spring guide 575 that are configured to move to compress the at least one spring 548 may cause the female spring guide 575 to remain in contact with or seated within the spring receiving portion 546 of the upper lever arm 534 and cause the male spring guide 577 to remain in contact with, or seated within, the spring retaining portion 547 formed in the lower arm 526. As such, regardless of the amount of travel of the pedal arm 502, there is a tension caused by the male spring guide 577 and the female spring guide 575 maintaining a position of contact of the spring carrier 570 extending between the upper lever arm 534 and the lower arm 526.

(110) As further depicted in FIGS. 5A-5F, it should be appreciated that the at least one spring 548 may include a plurality of spring coils. For example, in some embodiments, multiple spring coils may be utilized to ensure that the pedal emulator assembly 500 remains functional in the event at least one of the spring coils fails (e.g., breaks, disengages the spring retaining portion and/or spring retention portion, and the like) In these embodiments, the at least one spring 548 may be formed with a steel material, such as stainless steel, wire, carbon steel, alloy steel, elgiloy, Monel®, copper, nickel, and/or the like.

(111) Operation of the pedal emulator assembly 500 will now be discussed in detail with reference to FIGS. 5D-5F. For example, as depicted in FIG. 5D, the pedal emulator assembly 500 is an initial, or undepressed position, and/or subject to a first predetermined load L1. In the undepressed position, the link lever arm 520, which is guided by the pair of slots 518 formed in the housing 504 and are coupled to the pedal arm 502, as discussed in greater detail herein. In these embodiments, the at least one spring 548 may be in a mostly unbiased (e.g., fully extended) such that the at least one spring 548 is only applying a minimum resistance to maintain the positioning of the pedal arm 502. As such, the first predetermined load L1 may be enough force to compress the at least one spring 548 a predetermined distance and the elongated member 533 may slidably contact or slidably engage with a portion of the non-linear profile of the engagement surface 549 of the upper frame portion 538 of the upper lever arm 534 to achieve the desirable curve illustrated by bracket 102 in FIG. 1.

(112) Referring now to FIG. 5E, when a second predetermined load L2 is applied to the pedal pad 510, in the direction indicated by arrow A5, the pedal arm 502 moves or pivots about the pivot end 506, in the direction indicated by arrow A6, to allow the pad end 508 to move into and out of the housing 504 based on the amount of force supplied to the pad end 508 and the current position of the pedal arm 502. For example, as depicted in FIG. 5E, the pad end 508 may move partially into

the housing **504**, such that the at least one spring **548** is partially compressed and generates a return force or force feedback on the pedal pad **510** that corresponds to the amount of force applied to the pedal pad **510** by the driver.

(113) Simultaneously, the depression of the pedal arm **502** about the pivot end **506** may cause the slot **518** formed in the pedal arm **502** to contact the through pin that extends through the support end **522** of the link lever arm **520**, in addition to the compression of the at least one spring **548**. In these embodiments, the rotation of the pedal arm **502** about the pivot end **506** (e.g., in the direction indicated by arrow A2) and the contact between the slot **518** and the elongated member **533** may force the elongated member **533** to move in an upward direction along the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534**. As the elongated member **533** rides along the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534**, the link lever arm **520** may exert a downward force on the upper lever arm **534**, which may in turn cause the upper lever arm **534** to rotate about its second end **536b** and compress the at least one spring **548**.

(114) In these embodiments, the non-linear profile of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** may impact or affect the resistance of the pedal arm **502** experienced by the driver. For example, the contour of the engagement surface **549** of the upper frame portion **538** may impact the force that must be exerted on the pedal pad **510** by the driver in order to advance the through pin upwardly through the slot **518**, and in turn, compress the at least one spring **548**.

(115) More particularly, the non-linear contour of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** may generate a variable mechanical advantage between the link lever arm **520** and the upper lever arm **534**, which may allow for the pedal emulator assembly **500** to generate a desired force curve, as depicted in FIGS. 1 and 2. For example, as depicted most clearly in FIGS. 5D-5F, a first portion of the engagement surface **549** of the upper frame portion **538** (e.g., the portion of the engagement surface **549** that the elongated member **533** contacts in the undepressed position depicted in FIG. 5D) may include a relatively shallow, or gentle slope.

Because the slope of the first portion of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** is shallow, a second predetermined load L2, which may be a relatively low amount of force, may be required to depress the pedal pad **510** from the undepressed position (FIG. 5D) to a partially depressed position (FIG. 5E), which is depicted in bracket **104** of FIG. 1.

(116) As further depicted in FIGS. 5D-5F, a second portion of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** may have a slope that increases in steepness across the length of the second portion of the engagement surface **549** of the upper frame portion **538**. As the slope of the engagement surface **549** of the upper frame portion **538** increases, it may become increasingly difficult to depress the pedal pad **510**, such that a third predetermined load L3 that is greater than the second predetermined load L2 is required to move the elongated member **533** across the second portion of the engagement surface **549** of the upper frame portion **538** having the steeper slope. Because the force required to move the elongated member **533** through the slot **518** increases, the force exerted by the link lever arm **520** on the upper lever arm **534** is similarly increased, which may result in increased compression of the at least one spring **548**, as is depicted by bracket **106** of FIG. 1.

(117) In the embodiments described herein, it should be further appreciated that the elongated member **533** may not contact each portion of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534**, as the movement of the elongated member **533** along the engagement surface **549** of the upper frame portion **538** is restricted by the slot **518**. For example, the bottom surface and upper surface of the slot **518** may define the distance that the elongated member **533** may travel, and in turn, the portions of the engagement surface **549** of the upper frame portion **538** that the elongated member **533** may contact. Accordingly, it should be understood that the size of the slot may be configured to ensure that the elongated member **533** only contacts an

intended portion of the engagement surface **549** of the upper frame portion **538**. Furthermore, in these embodiments, the slot **518** ensures that the elongated member **533** rides along the engagement surface **549** of the upper frame portion **538** accurately. Accordingly, the guidance of the slot **518** helps ensure that the desired force curve of the pedal emulator assembly **500** is not disrupted.

(118) Referring now to FIG. 5F, the forces required to depress the pedal arm **502** may be greatest when the elongated member **533** contacts the portion of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** that has the greatest slope, which may correspond to a position when the pedal arm **502** is fully depressed. For example, when the pedal arm **502** is fully depressed, the at least one spring **548** may be fully compressed, such that the pedal arm **502** (and thus, the foot of the driver) may experience maximum resistance.

(119) Referring now to FIGS. 5A-5F, the pedal emulator assembly **500** may further include a sensor assembly **560** configured to monitor the rotational movement of the second end **536b** of the upper lever arm **534** to determine the exact force and position of the pedal pad **510** as the elongated member **533** of the primary lever arm **520** traverses in the slot **518**. In these embodiments, the sensor assembly **560** may further include a coupler **564** integrated into second end **536b** of the upper lever arm **534**, such that the coupler **564** may rotate in tandem with the second end **536b** of the upper lever arm **534** and convert mechanical movement (e.g., rotation) of the upper lever arm **534** into a measurable signal.

(120) In particular, rotation of the coupler **564** affects the orientation and position of the coupler **564** relative to a plurality of transmitting and receiving coils. For example, as the coupler **564** moves (e.g., rotates), it induces changes in a magnetic field emitted by the transmitting coil. The receiving coils detect these changes, and variations in the received magnetic field can then be converted into electrical signals, which are processed, by an onboard controller or microprocessor. Accordingly, the electrical signals indicative of the variations in the received magnetic field may be directly correlated to the rotational position of the upper lever arm **534**, and in turn, the pedal pad **510**.

(121) As should be appreciated in view of the foregoing, the various contours of the engagement surface **549** and the at least one spring **548** provide the pedal effect/resistance felt at the pedal pad **510** by the driver. That is, the force applied by the at least one spring **548** and the contours of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** apply different forces at the corresponding travel positions with the contours of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** near an end of the slot **518** (full travel of the pedal arm **502**) applying a significant force to the pedal arm **502**. The combination of the at least one spring **548** and the contours of the engagement surface **549** of the upper frame portion **538** of the upper lever arm **534** result in a composite of forces to provide a desired force curve.

(122) Turning now to FIGS. 6A-6D, a fourth example embodiment of a pedal emulator assembly **600** that utilizes a pivoting friction lever to generate resistance is schematically depicted. In these embodiments, the pedal emulator assembly **600** may be a floor mounted organ-type brake pedal emulator assembly. This is without limitation, and the pedal emulator assembly **600** may be a hanging pedal assembly or any other type of pedal assembly. The pedal emulator assembly **600** includes a pedal arm **602** and a housing **604**. The pedal arm **602** includes a pivot end **606** and an opposite pad end **608** in which a pedal pad **610** is attached thereto. The pedal arm **602** includes an exterior surface **612a** and an opposite interior surface **612b**. A link **614** is coupled to the interior surface **612b** and to a spring carrier assembly **616** positioned within a cavity **618** of the housing **604**.

(123) The spring carrier assembly **616** includes a lever arm **620** and a spring carrier **622**. The lever arm **620** includes a hub portion **624** and a spring receiving portion **626**. The spring carrier **622** includes a concave portion **628** and a spring retaining portion **630**.

(124) Referring still to FIGS. 6A-6D, the housing **604** includes a friction member **632** integrally

formed on an interior surface **634** of a sidewall **636**. The friction member **632** may be semi-annular or semi-circular. The hub portion **624** of the lever arm **620** abuts the friction member **632** of the housing **604** and the concave portion **628** of the spring carrier **622**. The hub portion **624** rotates within the concave portion **628** of the spring carrier **622** and against the friction member **632** of the housing **604** when a pressure is applied to the spring receiving portion **626**. The pressure is received when the pedal pad **610** is depressed, which drives the link **614** into the lever arm **620**. As such, pedal effort/resistance is generated by the friction created from the hub portion **624** rotating against the friction member **632** of the housing **604** and the concave portion **628** of the spring carrier **622**.

(125) In the embodiments described herein, the housing **604** and pedal arm **602**, may be a molded plastic. For example, the housing **604** and pedal arm **602** may be formed with various materials such as acrylonitrile butadiene styrene (ABS), polyethylene (PE), polypropylene (PP), polycarbonate (PC), nylon, polycarbonate/acrylonitrile butadiene styrene, polyurethane, polymethyl methacrylate, high density polyethylene, low density polyethylene, polystyrene, PEEK, POM (Acetal/Delrin), polyethylene terephthalate, thermoplastic elastomer, polyetherimide, thermoplastic vulcanizate, polysulfone, combinations thereof, and/or the like. Additionally, additives may be added such as UV absorbers, flame-retardants, colorants, glass fibers, plasticizers and/or the like.

(126) In other embodiments, the housing **604** and pedal arm **602** may be formed from injection molding or other additive manufacturing techniques. For example, as provided herein, additive manufacturing techniques refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present invention may use layer-additive processes, layer-subtractive processes, or hybrid processes.

(127) As further depicted in FIGS. **6A-6D**, the pedal emulator assembly **600** may further include a first compressible member, such as at least one first spring **638**, positioned between the spring receiving portion **626** of the lever arm **620** and the spring retaining portion **630** of the spring carrier **622** to bias the lever arm **620** in a direction opposite of a pedal depression force. The link **614** is coupled to the spring receiving portion **626**. When a pedal depression force is applied to the pedal arm **602**, the link **614** moves into the spring receiving portion **626** of the lever arm **620**, which in turn compresses the at least one first spring **638** into the spring retaining portion **630** of the spring carrier **622**.

(128) In these embodiments, the pedal emulator assembly **600** may further include a sensor assembly **652** that includes a coupler **654** mounted in the hub portion **624** and is configured to rotate when the hub portion **624** rotates. The sensor assembly **652** may track or determine the movement of the coupler **654** via a transmitting coil and at least one receiving coil positioned on or within a circuit board, such as a printed circuit board. As such, the sensor assembly **652** will sense the amount of force on the pedal pad **610** at any given time based on the position of the coupler **654**. Operation of the sensor assembly **652** will be described in additional detail herein.

(129) Referring still to FIGS. **6A-6D**, a second compressible member, such as at least one second spring **640**, is positioned at least partially within a spring receiving cavity **642** of the housing **604**. A portion of the at least one second spring **640** is positioned above a housing upper surface **644** and below the interior surface **612b** of the pedal arm **602**. As such, once the pedal pad **610** is depressed

with enough force to move the link **614** a predetermined distance, which is a force greater than the force needed to move the at least one first spring **638**, an upper surface **646** of the at least one second spring **640** makes contact with the interior surface **612b** of the pedal arm **602** compressing the at least one second spring **640** into the spring receiving cavity **642**. Such compression provides the pedal effort/resistance of the second section of the desired force curve of FIG. **1**, as will be described in additional herein.

(130) In these embodiments, the pedal emulator assembly **600** may further include an elastomer member **648** or foam member positioned on and extending from the housing upper surface **644** and below the interior surface **612b** of the pedal arm **602**. The elastomer member **648** is configured to compress at a predetermined rate when the interior surface **612b** of the pedal arm **602** makes contact with an upper surface **650** of the elastomer member **648**.

(131) Operation of the pedal emulator assembly **600** will now be described in detail with reference to FIGS. **6B-6D**. For example, as depicted in FIG. **6B**, when the pedal pad **610** receives a first predetermined load, the pedal pad **610** begins to pivot around its pivot end **606**. This movement is transferred via the link **614**, such that the link **614** moves the lever arm **620** which compresses the at least one first spring **638** and rotates the hub portion **624**. In these embodiments, the at least one first spring **638** is only applying a minimum resistance to maintain the positioning of the pedal arm **602**. A such, the first predetermined load **L1** may be enough force to compress the at least one first spring **638** a predetermined distance such that the link **614** contacts the spring receiving portion **626** of the spring carrier assembly **616** to achieve the desirable curve illustrated by bracket **102** in FIG. **1**.

(132) In these embodiments, as the first predetermined load **L1** acts on the pedal arm **602**, the hub portion **624** rotates within the concave portion **628** of the spring carrier **622**, such that the hub portion **624** contacts the friction member **632** of the housing **604**. Accordingly, the rotation of the hub portion **624** generates a frictional resistance, which acts as a portion of the first section of the pedal effort depicted as bracket **102** in the desired force curve depicted in FIG. **1**.

(133) Referring now to FIG. **6C**, when a second predetermined load **L2** that is larger than the first predetermined load **L1** acts on the pedal arm **602**, the link **614** continues to move (e.g., in the direction illustrated by arrow **A7**). In response, once the pedal arm **602** has moved a predetermined distance, the interior surface **612b** of the pedal arm **602** contacts the upper surface **646** of the at least one second spring **640**. In these embodiments, the force exerted on the pedal arm **602** is sufficient to overcome a combined resistance of the at least one first spring **638** and frictional forces acting between the hub portion **624** and the friction member **632**, such that the at least one second spring **640**, which is at least partially lodged in the spring receiving cavity **642**, begins to compress. As should be appreciated in view of the foregoing, the compression of the at least one second spring **640** may offer increased resistance to the pedal arm **602**. Accordingly, the second predetermined load **L2** may be enough force to compress both the at least one first spring **638** and the at least one second spring **640** a predetermined distance achieve the desirable curve illustrated by bracket **104** in FIG. **1**.

(134) Turning now to FIG. **6D**, a third predetermined load **L3** may be applied to the pedal arm **602** that is greater than the second predetermined load **L2** and the first predetermined load **L1** as the driver continues to exert force on the pedal pad **610**, such that the interior surface **612b** of the pedal arm comes into contact with the upper surface **650** of the elastomer member **648** positioned on the housing upper surface **644**. The elastomer member **648**, being made of a flexible material such as rubber, polyurethane, or any other similarly deformable material, may begin to compress under the force applied by the pedal arm **602** to the elastomer member **648**. In these embodiments, the compression of the elastomer member **648** may provide a significant resistance force, which may correspond to the third section of the desired force curve depicted by bracket **106** in FIG. **1**. For example, the resistance offered by the compression of the elastomer member **648** may be the highest resistance generated by the pedal emulator assembly **600**, which may indicate to a driver an

end-stop feeling of the pedal arm **602** depression.

(135) In view of the foregoing, it should be appreciated that the force applied by the various springs (e.g., the at least one first spring **638** and/or the at least one second spring **640**) and the elastomer member **648** apply different forces at corresponding travel positions. Compression of the elastomer member **648** applies a significant force to the pedal arm **602**. The combination of the various springs (e.g., the at least one first spring **638** and/or the at least one second spring **640**) and the elastomer member **648** result in a composite of forces to provide a desired force curve.

(136) Referring now to FIGS. **6B-6D** collectively, the sensor assembly **652** is configured to monitor the rotational movement of the hub portion **624** of the lever arm **620** to determine the exact force and position of the pedal pad **610** as the pedal arm **602** is depressed. In these embodiments, the coupler **654** may rotate in tandem with the hub portion **624** of the lever arm **620**, thereby converting mechanical movement (e.g., rotation) of the hub portion **624** into a measurable signal.

(137) In particular, rotation of the coupler **654** affects the orientation and position of the coupler **654** relative to the transmitting and receiving coils. For example, as the coupler **654** moves (e.g., rotates), it induces changes in a magnetic field emitted by the transmitting coil. The receiving coils detect these changes, and variations in the received magnetic field can then be converted into electrical signals, which are processed, by an onboard controller or microprocessor. Accordingly, the electrical signals indicative of the variations in the received magnetic field may be directly correlated to the rotational position of the hub portion **624**, and in turn, the pedal pad **610**.

(138) Now referring to FIGS. **7A-7D**, there is shown a schematic representation of a fifth embodiment of a pedal emulator assembly **700** that utilizes a rotational pedal arm, compression, and sliding friction to generate resistance. The pedal emulator assembly **700** includes a pedal arm **702** and a housing **704**. The pedal arm **702** is configured to rotate or pivot about a pivot hub **706** at a first end **708a** of the pedal arm **702**. An opposite second end **708b** includes a pedal pad **710** attached thereto. The pedal arm **702** includes an exterior surface **712a**, and an opposite interior surface **712b**. The housing **704** includes a friction member **714** integrally formed with and extending from an inner surface **716** of a sidewall **717**. The friction member **714** may be arcuate shaped and extend from a position below the interior surface **712b** of the pedal arm **702** to and partially around the first end **708a** or pivot hub **706**.

(139) The housing **704** further includes a spring receiving cavity **718** positioned below the interior surface **712b** of the pedal arm **702**. Further, the interior surface **712b** of the pedal arm includes a spring receiving portion **720** positioned between the first end **708a** and the second end **708b**. At least one compressible member, such as at least one spring **722**, extends between the spring receiving portion **720** and the spring receiving cavity **718** such that the at least one spring is compressible between the housing **704** and the pedal arm **702** when a force is applied to the pedal pad **710**.

(140) In the embodiments described herein, the housing **704** and pedal arm **702** may be a molded plastic. For example, the housing **704** and pedal arm **702** may be formed with various materials such as acrylonitrile butadiene styrene (ABS), polyethylene (PE), polypropylene (PP), polycarbonate (PC), nylon, polycarbonate/acrylonitrile butadiene styrene, polyurethane, polymethyl methacrylate, high density polyethylene, low density polyethylene, polystyrene, PEEK, POM (Acetal/Delrin), polyethylene terephthalate, thermoplastic elastomer, polyetherimide, thermoplastic vulcanizate, polysulfone, combinations thereof, and/or the like. Additionally, additives may be added such as UV absorbers, flame-retardants, colorants, glass fibers, plasticizers and/or the like.

(141) In other embodiments, the housing **704** and pedal arm **702** may be formed from injection molding or other additive manufacturing techniques. For example, as provided herein, additive manufacturing techniques refer generally to manufacturing processes wherein successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally fuse together to form a monolithic component which

may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling fabrication of complex objects by building objects point-by-point, layer-by-layer, typically in a vertical direction, other methods of fabrication are possible and within the scope of the present subject matter. For example, although the discussion herein refers to the addition of material to form successive layers, one skilled in the art will appreciate that the methods and structures disclosed herein may be practiced with any additive manufacturing technique or manufacturing technology. For example, embodiments of the present invention may use layer-additive processes, layer-subtractive processes, or hybrid processes.

(142) A friction member assembly **724** is coupled to the interior surface **712b** of the pedal arm **702** via a first link arm **726** between the spring receiving portion **720** and the first end **708a** to a friction shoe **728**. The first link arm **726** may be rotatably received within a semi-annular receiving portion **730** of the friction shoe **728** at or adjacent to a proximate end **732**. A second link arm **734** may be received within a receiving portion **736** at a distal end **738** of the friction shoe **728** and extend between the friction shoe **728** and the first end **708a** of the pedal arm **702**. As such, the friction member assembly **724** may include the friction shoe **728**, the first link arm **726**, and/or the second link arm **734**.

(143) The friction shoe **728** may further include a recess or notch **740** that receives a spring extending between the notch **740** and the second link arm **734** and/or a cantilever member **742** that is fixed or coupled to the second link arm **734** and moves within the notch **740** based on the amount of rotation of the first end **708a** about the pivot hub **706**. The friction shoe **728** abuts or rides within the friction member **714**, which is arcuate, or non-linear. That is, the friction shoe **728** may be arcuate shaped or non-linear.

(144) Operation of the pedal emulator assembly will now be described in detail with reference to FIGS. 7B-7D. As depicted in FIG. 7B, when a first predetermined load **L1** is applied to the pedal pad **710**, the first end **708a** and the pedal arm begins to rotate about the pivot hub **706**. In these embodiments, rotation of the pedal arm **702** about the pivot hub **706** causes the interior surface **712b** of the pedal pad **710** to contact and compress the at least one spring **722**. In these embodiments, the at least one spring **722** may be in a mostly unbiased position, such that the at least one spring **722** is only applying a minimum resistance to maintain the position of the pedal arm **702**. As such, the first predetermined load **L1** may be enough force to compress the at least one spring **722** a predetermined distance, such that the spring receiving portion **720** of the pedal arm **702** contacts the at least one spring **722** to achieve the desirable force curve illustrated by bracket **102** of FIG. 1

(145) Simultaneously, the rotation of the pedal arm **702** actuates the first link arm **726**, which, in turn, pushes the friction shoe **728** along the friction member **714**. In these embodiments, the shape and/or material of the friction shoe **728** and friction member **714** may allow for the introduction of sliding friction into the pedal emulator assembly **700**, which forms a portion of the resistance generated by the first predetermined load **L1** and as depicted in the bracket **102** of FIG. 1. For example, and as should be appreciated in view of the foregoing, higher frictional forces may be achieved between the friction shoe **728** and the friction member **714** by increasing the contact surface area between the two components.

(146) In the embodiment depicted in FIGS. 7A-7D, as the friction shoe **728** contacts various portions of the friction member **714**, the angle of contact between the friction shoe **728** and the friction member **714** may change due to the non-linear, or arcuate, profile of the friction member **714**. Accordingly, when the friction shoe **728** contacts portions of the friction member **714** having a steeper curvature, the friction shoe **728** may produce a larger normal force (e.g., a force perpendicular to the surface of the friction member **714**), which may in turn result in an increased frictional force.

(147) Referring again to FIG. 7B, as the pedal pad **710** receives the first predetermined load **L1**, the first link arm **726** and the second link arm **734** may act as connectors between the pedal arm **702**

and the friction shoe **728**. Accordingly, in these embodiments, the first link arm **726** and the second link arm **734** may translate the rotational motion of the pedal arm **702** into linear motion of the friction shoe **728**. In these embodiments, the non-linear profile of the friction member **714** may result in the generation of frictional force between the friction shoe **728** and the friction member **714** as the friction shoe is linearly actuated, as has been described in detail herein.

(148) Turning now to FIG. 7C, as the pedal arm **702** continues to rotate about the pivot hub **706**, the pedal arm **702** may receive a second predetermined load **L2**. In these embodiments, when the second predetermined load **L2** is applied to the pedal arm, the interior surface **712b** of the pedal arm **702** may further compress the at least one spring **722**, thereby increasing resistance within the pedal emulator assembly **700** and making the pedal arm **702** more difficult to depress (e.g., rotate). Furthermore, as the pedal arm **702** rotates, the friction shoe **728** continues to traverse the friction member **714**. As the friction shoe **728** interacts with a portion of the friction member **714** having a steeper curve, the resistance due to friction increases, as has been described herein. Accordingly, it should be appreciated that the progressive increase in friction resulting from the non-linear profile of the friction member **714** may simulate the braking force in a traditional brake pad and provide tactile feedback to the driver.

(149) Referring still to FIG. 7C, as the second predetermined load **L2** acts on the pedal arm **702**, the position of the cantilever member **742** within the notch **740** may also change. For example, and as described herein, rotation of the pedal arm **702** may cause the first link arm **726** to move the friction shoe **728** along the friction member **714**. As the friction shoe **728** moves, the cantilever member **742** may be subject to displacement due to the connection between the cantilever member **742** and the second link arm **734**. Accordingly, as the pedal arm **702** continues to rotate, the cantilever member **742** may become increasingly displaced, which may cause the cantilever member **742** to bend. In these embodiments, the bending of the cantilever member **742** and the connection between the cantilever member **742** and the second link arm **734** may provide a leverage effect that exerts a force on the second link arm **734**. More particularly, the force exerted by the cantilever member **742** against the second link arm is directed towards the pivot hub **706** of the pedal arm **702**, thereby opposing the rotation of the pedal arm **702**. Accordingly, the force exerted by the cantilever member **742** on the second link arm **734** may further contribute to the resistance of the pedal pad **710** and the tactile feedback experienced by the driver.

(150) In the embodiments described herein, it should be appreciated that the resistance generated by the cantilever member **742**, the friction member **714**, and the at least one spring **722** when the second predetermined load **L2** acts on the pedal arm **702** may produce a greater pedal effect experienced by the driver, as is illustrated in bracket **104** of FIG. 1

(151) Turning now to FIG. 7D, the pedal emulator assembly **700** is depicted in a fully compressed position, in which a third predetermined load **L3** that is greater than the first predetermined load **L1** and the second predetermined load **L2** is being applied to the pedal arm. In these embodiments, the pedal arm **702** may be fully rotated, such that the interior surface **712b** of the pedal pad **710** fully compresses the at least one spring **722**. Furthermore, the friction shoe **728** may interact with a portion of the friction member **714** having the most aggressive curve (e.g., steepest slope), which results in a maximum level of frictional force being exerted on the pedal pad **710**. Similarly, the first link arm **726** actuates the friction shoe **728** at a maximum distance along the friction member **714**, while the cantilever member **742** exerts a maximum force on the second link arm **734**. In these embodiments, the combination of the of forces may result in a peak resistance being achieved within the pedal emulator assembly **700**, which may correspond to the third section depicted as bracket **106** in the desired force curve illustrated in FIG. 1.

(152) In view of the foregoing, it should be appreciated that the force applied by the at least one spring **722**, friction shoe **728**, the curvature of the friction member **714**, and the various link arms **726**, **734** apply different forces at the corresponding travel positions. Compression of the second link arm **734** and curvature of the friction member **714** with the friction shoe **728** at that particular

point applies a significant force to the pedal arm **702**. The combination of the at least one spring **722**, friction shoe **728**, curvature of the friction member **714** and the various link arms **726**, **734** result in a composite of forces to provide a desired force curve.

(153) Referring again to FIGS. 7A-7D, the pedal emulator assembly **700** may further include a sensor assembly **750** is configured to monitor the rotational movement of the hub portion **624** of the pivot hub **706** to determine the exact force and position of the pedal pad **710** as the pedal arm **702** is depressed. In these embodiments, the sensor assembly **750** may further include a coupler **754** that may be integrated into the pivot hub **706**, such that the coupler **756** may rotate in tandem with the pivot hub **706** of the pedal arm **702** and convert mechanical movement (e.g., rotation) of the pivot hub **706** into a measurable signal.

(154) In particular, rotation of the coupler **754** affects the orientation and position of the coupler **754** relative to a plurality of transmitting and receiving coils (not depicted). For example, as the coupler **754** moves (e.g., rotates), it induces changes in a magnetic field emitted by the transmitting coil. The receiving coils detect these changes, and variations in the received magnetic field can then be converted into electrical signals, which are processed, by an onboard controller or microprocessor. Accordingly, the electrical signals indicative of the variations in the received magnetic field may be directly correlated to the rotational position of the pivot hub **706**, and in turn, the pedal pad **710**.

1. A pedal emulator assembly comprising: a housing having a cavity defined by a pair of sidewalls, a first end wall and a second end wall; a lower arm having a spring retaining portion formed therein; a pedal arm pivotally coupled the pedal arm and having a pedal pad; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further including a spring receiving portion; and a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion; wherein, when a first predetermined load is applied to the pedal pad, the pedal arm drives the lever arm into the compressible member such that the compressible member compresses into an at least partially compressed state to generate a first return force on the pedal pad. 2. The pedal emulator assembly of any preceding clause, wherein the lever arm further includes an upper frame portion that includes an engagement surface having a non-linear profile that extends between a first end of the lever arm and a second end of the lever arm. 3. The pedal emulator assembly of any preceding clause, wherein the upper frame portion includes a first portion extending between the first end of the lever arm and an apex of the lever arm. 4. The pedal emulator assembly of any preceding clause, wherein the upper frame portion includes a second portion extending between the apex of the lever arm and the second end of the lever arm. 5. The pedal emulator assembly of any preceding clause, wherein the first portion of the upper frame portion has a first slope and the second portion of the upper frame portion has a second slope different from the first slope. 6. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal pad, the pedal arm further contacts the first portion of the upper frame portion to generate a second return force on the pedal pad, the second predetermined load is a greater load than the first predetermined load and the second return force is a greater force than the first return force. 7. The pedal emulator assembly of any preceding clause, wherein, when a third predetermined load is applied to the pedal pad, the pedal arm further contacts the second portion of the upper frame to generate a third return force on the pedal pad, the third predetermined load is a greater load than the second predetermined load, and the third return force is a greater force than the second return force. 8. The pedal emulator assembly of any preceding clause, wherein the lower arm is fixedly coupled to within the cavity of the housing. 9. The pedal emulator assembly of any preceding clause, wherein the lower arm further includes a concave portion configured to receive a pivot portion of the pivot end of the pedal arm, such that the pedal arm pivots about the pivot portion within the concave portion of the lower arm. 10. The pedal emulator assembly of any preceding clause, wherein the compressible member includes a plurality of springs coaxially aligned about the spring carrier. 11. The pedal emulator assembly of any preceding clause, further

comprising: a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein the sensor assembly is configured to detect movement of the pedal arm. 12. A pedal emulator assembly for a vehicle comprising: a floor mounted housing having a cavity defined by a pair of side walls, a first end wall, and a second end wall; a lower arm having a spring retaining portion formed therein; a pedal arm at least partially received in the cavity, the pedal arm having an interior surface, a pivot end, and an opposite pad end on which a pedal pad is attached; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further comprising: a first end; a second end positioned opposite the first end; an apex disposed at the first end; a hub portion disposed at the second end; an upper frame portion and a lower frame portion that connect at the apex and the hub portion; and a spring receiving portion positioned between the first end and the second end and within the lower frame portion; and a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion; wherein, when a first predetermined load is applied to the pedal pad, the interior surface of the pedal arm drives the lever arm into the compressible member such that the compressible member compresses into a compressed state to generate a first return force on the pedal pad. 13. The pedal emulator assembly of any preceding clause, wherein the interior surface of the pedal arm is non-linear. 14. The pedal emulator assembly of any preceding clause, wherein the upper frame portion includes a surface having a non-linear profile that extends between the first end of the lever arm and the second end of the lever arm. 15. The pedal emulator assembly of any preceding clause, wherein the upper frame portion includes a first portion extending between the first end of the lever arm and the apex of the lever arm and a second portion extending between the apex of the lever arm and the second end of the lever arm. 16. The pedal emulator assembly of any preceding clause, wherein the first portion of the upper frame portion has a first slope and the second portion of the upper frame has a second slope different from the first slope. 17. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal pad, the interior surface of the pedal arm contacts the first portion of the upper frame portion to generate a second return force on the pedal pad, the second predetermined load is a greater load than the first predetermined load, and the second return force is a greater force than the first return force. 18. The pedal emulator assembly of any preceding clause, wherein, when a third predetermined load is applied to the pedal pad, the interior surface of the pedal arm contacts the second portion of the upper frame portion to generate a third return force on the pedal pad, the third predetermined load is a greater load than the second predetermined load, and the third return force is a greater force than the second return force. 19. The pedal emulator assembly of any preceding clause, further comprising: a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein the sensor assembly is configured to detect movement of the pedal arm. 20. A pedal emulator assembly for a vehicle comprising: a housing having a cavity defined by a pair of side walls, a first end wall and a second end wall; a lower arm having a spring retaining portion formed therein; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end pivotally coupled to the housing, an opposite pad end on which a pedal pad is attached, and an interior surface that is non-linear; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further comprising: a first end; a second end positioned opposite the first end; an apex disposed at the first end; a hub portion disposed at the second end; an upper frame and a lower frame that connect at the apex and the hub portion, the upper frame including an upper surface with a non-linear profile; and a spring receiving portion positioned between the first end and the second end in the lower frame; a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion; and wherein, when a first predetermined

load is applied to the pedal pad, the interior surface of the pedal arm contacts the upper surface of the upper frame of the lever arm, such that the lever arm is forced into the compressible member to generate a return force on the pedal pad. 21. A pedal emulator assembly comprising: a housing including a cavity; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end, an opposite pad end on which a pedal pad is attached, and a protrusion extending from an interior surface of the pedal arm; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further including an engagement surface configured to engage a contact surface of the protrusion; and a spring carrier including a compressible member extending between the pivot end of the pedal arm and the lever arm; wherein, when a first predetermined load is applied to the pedal pad, the contact surface of the protrusion of the pedal arm slidably engages against the engagement surface of the lever arm and generates a first return force on the pedal pad. 22. The pedal emulator assembly of any preceding clause, wherein the outer surface of the lever arm is a non-linear surface. 23. The pedal emulator assembly of any preceding clause, wherein the engagement surface of the housing is a non-linear surface. 24. The pedal emulator assembly of any preceding clause, wherein the engagement surface of the lever arm and the contact surface of the protrusion are configured such that, when a second predetermined load is applied to the pedal arm, contact between the engagement surface of the lever arm and the contact surface of the projection increases. 25. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal arm, rotation of the pedal arm compresses the compressible member to generate a second return force on the pedal pad. 26. The pedal emulator assembly of any preceding clause, further comprising an elastomer member positioned within a recess formed in the housing. 27. The pedal emulator assembly of any preceding clause, wherein an interior surface of the pedal arm at the pad end contacts and compresses the elastomer member when a third predetermined load is applied to the pedal pad. 28. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein the sensor assembly is configured to detect movement of the pedal arm. 29. The pedal emulator assembly of any preceding clause, wherein the housing and the pedal arm are formed of molded plastic. 30. A pedal emulator assembly comprising: a housing including a cavity; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end, an opposite pad end on which a pedal pad is attached, and a protrusion extending from an interior surface of the pedal arm; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further including an outer surface configured to engage a contact surface of the protrusion; and a spring carrier having a pedal end and an opposite lever arm end, the spring carrier extending between the pivot end of the pedal arm and the lever arm; and an elastomer member positioned within a recess formed in the housing; wherein, when a first predetermined load is exerted on the pedal pad, the contact surface of the protrusion of the pedal arm slides against an engagement surface of the lever arm and generates first return force on the pedal pad. 31. The pedal emulator assembly of any preceding clause, wherein the outer surface of the lever arm is a non-linear surface. 32. The pedal emulator assembly of any preceding clause, wherein the engagement surface of the housing is a non-linear surface. 33. The pedal emulator assembly of any preceding clause, wherein the interior surface of the pedal arm and the outer surface of the lever arm are configured such that contact between the outer surface of the lever arm and the interior surface of the pedal arm increases as the pedal pad is depressed from an initial position to a fully depressed position. 34. The pedal emulator assembly of any preceding clause, wherein the interior surface of the pedal arm at the pad end contacts and compresses the elastomer member when a third predetermined load is applied to the pedal pad. 35. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip

communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein the sensor assembly is configured to detect movement of the pedal arm. 36. The pedal emulator assembly of any preceding clause, wherein the pedal arm and the housing are formed from molded plastic. 37. The pedal emulator assembly of any preceding clause, wherein the lever arm further includes a protrusion that engages a recess formed in the cavity of the housing. 38. The pedal emulator assembly of any preceding clause, wherein the elastomer member is formed of rubber or plastic. 39. The pedal emulator assembly of any preceding clause, wherein the housing is configured to be mounted to an instrument panel. 40. A pedal emulator assembly for a vehicle comprising: a housing including a cavity; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end, an opposite pad end on which a pedal pad is attached, and a protrusion extending from an interior surface of the pedal arm; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further including an engagement surface configured to engage a contact surface of the protrusion of the pedal arm; and a spring carrier having a pedal end and an opposite lever arm end, the spring carrier extending between the pivot end of the pedal arm and the lever arm; an elastomer member positioned within a recess formed in the housing; and a sensor assembly positioned within the housing and configured to detect movement of the pedal arm, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein, when a first predetermined load is applied to the pedal pad, the interior surface of the pedal arm slides against the engagement surface of the lever arm and generates frictional force on the pedal pad. 41. A pedal emulator assembly comprising: a housing having a cavity defined by a pair of opposing side walls; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached; a pair of slots formed in the housing, the pair of slots extending through the pair of opposing side walls of the cavity; a lower arm having a first end pivotally coupled to the pivot end of the pedal arm and an opposite second end, the lower arm including a spring retaining portion; a link lever arm including a support end having an aperture extending there through, and a coupled end that is pivotally coupled to the lower arm; an upper lever arm pivotally coupled to the housing, the upper lever arm including a spring receiving portion; an elongated member extending through the pair of slots and the aperture of the link lever arm and contacting the upper lever arm; and a compressible member extending between the spring receiving portion of the upper lever arm and the spring retaining portion of the lower arm; wherein, when a first predetermined load is applied to the pedal pad, contact between the pair of slots and the elongated member drives the elongated member along the upper lever arm, such that the upper lever arm is forced into contact with the compressible member and compresses the compressible member to generate a first return force on the pedal pad. 42. The pedal emulator assembly of any preceding clause, wherein the upper lever arm includes an upper frame having an engagement surface having a non-linear profile that extends between a first end of the upper lever arm and an apex of the upper frame of the upper lever arm. 43. The pedal emulator assembly of any preceding clause, wherein the engagement surface includes a first portion extending partially between the first end of the upper lever arm and the apex of the upper lever arm. 44. The pedal emulator assembly of any preceding clause, wherein the first portion of the engagement surface has a first slope that changes along a length of the first portion of the upper frame portion. 45. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal pad, the elongated member contacts the first portion of the engagement surface to generate a second return force on the pedal pad. 46. The pedal emulator assembly of any preceding clause, wherein the engagement surface includes a second portion extending between the first portion of the engagement surface and the apex of the upper lever arm. 47. The pedal emulator assembly of any preceding clause, wherein the second portion of the engagement surface has a second slope that increases in steepness along a length of the second portion of the upper frame. 48. The pedal

emulator assembly of any preceding clause, wherein, when a third predetermined load is applied to the pedal pad, the elongated member contacts the second portion of the engagement surface to generate a third return force on the pedal pad. 49. The pedal emulator assembly of any preceding clause, wherein the elongated member further includes a pair of terminating ends having a first diameter that is larger than a second diameter of each of the pair of slots. 50. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of a second end of the upper lever arm. 51. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the second end of the upper lever arm, such that the coupler rotates in tandem with the second end of the upper lever arm. 52. The pedal emulator assembly of any preceding clause, wherein the housing and the pedal arm are formed of molded plastic. 53. The pedal emulator assembly of any preceding clause, wherein the compressible member comprises a spring extending between the spring receiving portion of the upper lever arm and the spring retaining portion of the lower arm. 54. A pedal emulator assembly for a vehicle comprising: a floor mounted housing having a cavity defined by a pair of opposing side walls; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached; a pair of slots formed in the housing, the pair of slots extending through the pair of opposing side walls of the cavity; a lower arm having a first end pivotally coupled to the pivot end of the pedal arm and an opposite second end, the lower arm including a spring retaining portion; a link lever arm including a support end having an aperture extending there through, and a coupled end that is pivotally coupled to the lower arm; an upper lever arm pivotally coupled to the housing, the upper lever arm comprising: a first end and second end positioned opposite the first end; an apex disposed at the first end; a hub portion disposed at the second end; an upper frame and a lower frame that connect at the apex and the hub portion, the upper frame including an engagement surface with a non-linear profile; and a spring receiving portion positioned between the first end and the second end in the lower frame; an elongated member extending through the pair of slots and the aperture of the link lever arm and contacting the upper lever arm; and a compressible member extending between the spring receiving portion of the upper lever arm and the spring retaining portion of the lower arm; wherein, when a first predetermined load is applied to the pedal pad, contact between the pair of slots and the elongated member drives the elongated member along the engagement surface of the upper lever arm, such that the lower frame of the upper lever arm is forced into contact with the compressible member and compresses the compressible member to generate a first return force on the pedal pad. 55. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the second end of the upper lever arm. 56. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the second end of the upper lever arm, such that the coupler rotates in tandem with the second end of the upper lever arm. 57. The pedal emulator assembly of any preceding clause, wherein the support end of the link lever arm includes a pair of ears, each of the pair of ears including the aperture for receiving the elongated member. 58. The pedal emulator assembly of any preceding clause, wherein the compressible member comprises a spring having a plurality of coils that each extend between the spring receiving portion of the upper lever arm and the spring retaining portion of the lower arm. 59. The pedal emulator assembly of any preceding clause, wherein a slope of the engagement surface of the upper frame increases in steepness along a length from the first end to the apex of the upper frame. 60. A pedal emulator assembly for a vehicle comprising: a floor mounted housing having a cavity defined by a pair of opposing side walls; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached; a pair of slots formed in the housing, the pair of slots extending through the pair of opposing side walls of the cavity; a lower arm having a first end pivotally coupled to the pivot end of the pedal arm and an opposite second end, the lower arm including a spring retaining portion; a link lever arm including a support end having a pair of ears,

each of the pair of ears having an aperture extending there through, and a coupled end that is pivotally coupled to the lower arm; an upper lever arm pivotally coupled to the housing, the upper lever arm comprising: a first end and second end positioned opposite the first end; an apex disposed at the first end; a hub portion disposed at the second end; an upper frame and a lower frame that connect at the apex and the hub portion, the upper frame including an engagement surface having a slope that increases in steepness from the first end to the apex of the upper lever arm; and a spring receiving portion positioned between the first end and the second end in the lower frame; an elongated member extending through the pair of slots and the aperture of the link lever arm and contacting the engagement surface of the upper lever arm; and a compressible member extending between the spring receiving portion of the upper lever arm and the spring retaining portion of the lower arm; wherein, when a first predetermined load is applied to the pedal pad, contact between the pair of slots and the elongated member drives the elongated member along the engagement surface of the upper lever arm, such that the lower frame of the upper lever arm is forced into contact with the compressible member and compresses the compressible member to generate a first return force on the pedal pad. 61. A pedal emulator assembly comprising: a housing having a cavity and a friction member formed on an interior surface of the housing; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached, a spring carrier assembly comprising: a lever arm having a spring receiving portion and a hub portion that abuts the friction member of the housing; and a spring carrier including a spring retaining portion; a link coupled to the pedal arm and the lever arm of the spring carrier assembly; and a compressible member positioned between the spring receiving portion of the lever arm and the spring retaining portion of the spring carrier; wherein, when a first predetermined load is applied to the pedal pad, the link drives the lever arm into the compressible member such that the compressible member compresses and the hub portion rotates against the friction member to generate a first return force on the pedal pad. 62. The pedal emulator assembly of any preceding clause, wherein the housing is mounted to a floor of a vehicle. 63. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing has a semi-annular or semi-circular profile. 64. The pedal emulator assembly of any preceding clause, wherein the housing further includes a spring receiving cavity. 65. The pedal emulator assembly of any preceding clause, wherein a second compressible member is positioned at least partially within the spring receiving cavity. 66. The pedal emulator assembly of any preceding clause, wherein, when distance second predetermined load is applied to the pedal pad, the pedal arm contacts the second compressible member and compresses the second compressible member into the spring receiving cavity. 67. The pedal emulator assembly of any preceding clause, further comprising an elastomer member extending from the housing. 68. The pedal emulator assembly of any preceding clause, wherein, when distance third predetermined load is applied to the pedal pad, the pedal arm contacts the elastomer member and compresses the elastomer member against the housing. 69. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the lever arm. 70. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the hub portion of the lever arm, such that the coupler rotates in tandem with the hub portion of the lever arm. 71. A pedal emulator assembly comprising: a housing having a cavity, a spring receiving cavity, and a friction member formed on an interior surface of the housing; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached; a spring carrier assembly comprising: a lever arm having a spring receiving portion and a hub portion that abuts the friction member of the housing; and a spring carrier including a spring retaining portion; a link coupled to the pedal arm and the lever arm of the spring carrier assembly; a first compressible member positioned between the spring receiving portion of the lever arm and the spring retaining portion of the spring carrier; and a second compressible member is positioned at least partially within the spring receiving cavity, wherein, when a first predetermined load is

applied to the pedal pad, the link drives the lever arm into the first compressible member, and when a second predetermined load is applied to the pedal pad, the link drives the lever arm into the first compressible member and the second compressible member, such that the first compressible member and the second compressible member compress and the hub portion rotates against the friction member to generate a second return force on the pedal pad. 72. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing has a semi-annular or semi-circular profile. 73. The pedal emulator assembly of any preceding clause, further comprising an elastomer member extending from the housing. 74. The pedal emulator assembly of any preceding clause, wherein, when a third predetermined load is applied to the pedal pad, the pedal arm contacts the elastomer member and compressed the elastomer member against the housing. 75. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the lever arm. 76. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the hub portion of the lever arm, such that the coupler rotates in tandem with the hub portion of the lever arm. 77. A pedal emulator assembly comprising: a housing having a cavity, a spring receiving cavity, and a friction member formed on an interior surface of the housing; a pedal arm at least partially received in the cavity, the pedal arm having a pivot end and an opposite pad end on which a pedal pad is attached; a spring carrier assembly comprising: a lever arm having a spring receiving portion and a hub portion that abuts the friction member of the housing; and a spring carrier including a spring retaining portion; a link coupled to the pedal arm and the lever arm of the spring carrier assembly; a first compressible member positioned between the spring receiving portion of the lever arm and the spring retaining portion of the spring carrier; a second compressible member is positioned at least partially within the spring receiving cavity; and an elastomer member extending from the housing; wherein, when a first predetermined load is applied to the pedal pad, the pedal arm rotates such that the link drives the lever arm into the first compressible member, and when a second predetermined load is applied to the pedal pad, the link drives the lever arm into the second compressible member, and when a third predetermined load is applied to the pedal pad, an interior surface of the pedal arm compresses the elastomer member to generate a return force on the pedal pad. 78. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing has a semi-annular or semi-circular profile. 79. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the lever arm. 80. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the hub portion of the lever arm, such that the coupler rotates in tandem with the hub portion of the lever arm. 81. A pedal emulator assembly comprising: a housing having a spring receiving cavity and a friction member formed on an inner surface of the housing; a pedal arm at least partially received in the housing, the pedal arm having a pivot end an opposite pad end on which a pedal pad is attached, and a spring receiving portion positioned between the pivot end and the opposite pad end on which the pedal pad is attached; a compressible member extending between the spring receiving cavity and the spring receiving portion of the pedal arm; and a friction member assembly comprising: a friction shoe that abuts the friction member of the housing; a first link arm extending between the spring receiving portion of the pedal arm and a proximate end of the friction shoe; and a second link arm extending between the pivot end of the pedal arm and a distal end of the friction shoe; wherein, when a first predetermined load is applied to the pedal pad, the pedal arm compresses the compressible member to generate a first return force on the pedal pad. 82. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal pad, the friction shoe rotates against the friction member to generate a second return force on the pedal pad. 83. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing is arcuate shaped. 84. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing extends from a position below an

interior surface of the pedal arm to and partially around the pivot end of the pedal arm. 85. The pedal emulator assembly of any preceding clause, wherein the friction shoe is arcuate shaped. 86. The pedal emulator assembly of any preceding clause, wherein the friction shoe further comprises a notch. 87. The pedal emulator assembly of any preceding clause, wherein the friction assembly further includes a cantilever member coupled to the second link arm and extending between the second link arm and the notch of the friction shoe, such that the cantilever member moves within the notch. 88. The pedal emulator assembly of any preceding clause, wherein, when a second predetermined load is applied to the pedal pad, the cantilever member is displaced within the notch, such that the cantilever member bends and exerts a force on the second link arm. 89. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the pedal pad. 90. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the pivot end of the pedal arm, such that the coupler rotates in tandem with the pivot end of the pedal arm. 91. A pedal emulator assembly comprising: a housing having a spring receiving cavity and a friction member formed on an inner surface of the housing; a pedal arm at least partially received in the housing, the pedal arm having a pivot end an opposite pad end on which a pedal pad is attached, and a spring receiving portion positioned between the pivot end and the opposite pad end on which the pedal pad is attached; a compressible member extending between the spring receiving cavity and the spring receiving portion of the pedal arm; and a friction assembly comprising: a friction shoe that abuts the friction member of the housing; a first link arm extending between the spring receiving portion of the pedal arm and a proximate end of the friction shoe; a second link arm extending between the pivot end of the pedal arm and a distal end of the friction shoe; a notch; and a cantilever member coupled to the second link arm and extending between the second link arm and the notch of the friction shoe, such that the cantilever member moves within the notch; wherein, when a first predetermined load is applied to the pedal pad, the pedal arm compresses the compressible member, and when a second predetermined load is applied to the pedal pad, the friction shoe rotates against the friction member to generate a return force on the pedal pad. 92. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing is arcuate shaped. 93. The pedal emulator assembly of any preceding clause, wherein the friction member of the housing extends from a position below an interior surface of the pedal arm to and partially around the pivot end of the pedal arm. 94. The pedal emulator assembly of any preceding clause, wherein the friction shoe is arcuate shaped. 95. The pedal emulator assembly of any preceding clause, wherein, when a third predetermined load is applied to the pedal pad, the cantilever member is displaced within the notch, such that the cantilever member bends and exerts a force on the second link arm. 96. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the pedal pad. 97. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the pivot end of the pedal arm, such that the coupler rotates in tandem with the pivot end of the pedal arm. 98. A pedal emulator assembly for a vehicle comprising: a floor mounted housing having a spring receiving cavity and an arcuate shaped friction member formed on an inner surface of the housing; a pedal arm at least partially received in the housing, the pedal arm having a pivot end an opposite pad end on which a pedal pad is attached, and a spring receiving portion positioned between the pivot end and the opposite pad end on which the pedal pad is attached; a compressible member extending between the spring receiving cavity and the spring receiving portion of the pedal arm; and a friction assembly comprising: an arcuate shaped friction shoe that abuts the friction member of the housing; a first link arm extending between the spring receiving portion of the pedal arm and a proximate end of the friction shoe; a second link arm extending between the pivot end of the pedal arm and a distal end of the friction shoe; a notch; and a cantilever member coupled to the second link arm and extending between the second link arm and the notch of the friction shoe, such that the cantilever member moves within the notch;

wherein, when a first predetermined load is applied to the pedal arm, the pedal arm compresses the compressible member, when a second predetermined load is applied to the pedal arm, the friction shoe rotates against the friction member, and when a third predetermined load is applied to the pedal arm, the cantilever member is displaced within the notch to generate a return force on the pedal pad. 99. The pedal emulator assembly of any preceding clause, further comprising a sensor assembly configured to monitor rotational movement of the pedal pad. 100. The pedal emulator assembly of any preceding clause, wherein the sensor assembly further comprises a coupler integrated into the pivot end of the pedal arm, such that the coupler rotates in tandem with the pivot end of the pedal arm.

(155) It should now be understood that the embodiments described herein are directed to embodiments of pedal emulator assemblies that include sensors adapted to sense a position and/or force of a pedal pad. The various pedal assemblies are configured to simulate a braking fluid system, such as a hydraulic system, that is based on a speed system such as the faster a driver depresses onto the pedal pad of a pedal, the harder or more difficult the pedal is to depress, known herein as hepatic force. The pedal emulator assemblies described herein apply an opposite emulator force to provide the driver with a resistive force that changes according to the speed in which the PE is applied. Generally, the emulator assembly outputs three distinct force vs travel sections that correspond to a specific range of travel such that the driver feels different resistance or haptics based on how fast the pedal is being depressed.

(156) The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms, including “at least one,” unless the content clearly indicates otherwise. “Or” means “and/or.” As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. The term “or a combination thereof” means a combination including at least one of the foregoing elements.

(157) It is noted that the terms “substantially” and “about” may be utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. These terms are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

(158) While particular embodiments have been illustrated and described herein, it should be understood that various other changes and modifications may be made without departing from the spirit and scope of the claimed subject matter. Moreover, although various aspects of the claimed subject matter have been described herein, such aspects need not be utilized in combination. It is therefore intended that the appended claims cover all such changes and modifications that are within the scope of the claimed subject matter.

Claims

1. A pedal emulator assembly comprising: a housing defined by a pair of sidewalls, a first end wall and a second end wall; a pedal arm having a pedal pad and having a pair of opposing walls, the pedal arm pivotally coupled the housing such that the pair of opposing walls are movable relative to the pair of sidewalls thereby defining a cavity by the pair of opposing walls and the housing; a lower arm having a spring retaining portion formed therein; a lever arm pivotally coupled to the housing, the lever arm further including: a spring receiving portion; and an engagement surface configured to slidably engage with a portion of the pedal arm, the engagement surface having a

non-linear profile that extends a length between a first end positioned at a distal end of the lever arm and a second end of the lever arm, the non-linear profile configured to provide a different force characteristic based on an engagement position of contact between the portion of the pedal arm and the engagement surface substantially along the length of the non-linear profile based on a load applied to the pedal pad, the engagement surface is positioned to be enclosed within the cavity; and a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion, wherein, when the load is a first predetermined load applied to the pedal pad, the pedal arm drives the lever arm into the compressible member such that the compressible member compresses into an at least partially compressed state to generate a first return force on the pedal pad.

2. The pedal emulator assembly of claim 1, wherein the engagement surface of the lever arm is defined by an upper frame portion.

3. The pedal emulator assembly of claim 2, wherein the upper frame portion includes a first portion extending between the first end of the lever arm and an apex of the engagement surface of the lever arm.

4. The pedal emulator assembly of claim 3, wherein the upper frame portion includes a second portion extending between the apex of the engagement surface of the lever arm and the second end of the lever arm.

5. The pedal emulator assembly of claim 4, wherein the engagement surface of the first portion of the upper frame portion has a first slope and the engagement surface of the second portion of the upper frame portion has a second slope different from the first slope.

6. The pedal emulator assembly of claim 1, wherein, when the load is a second predetermined load applied to the pedal pad, the pedal arm further contacts a different engagement position of contact of the engagement surface than the engagement position of contact when the load is the first predetermined load to generate a second return force on the pedal pad, the second predetermined load is a greater load than the first predetermined load and the second return force is a greater force than the first return force.

7. The pedal emulator assembly of claim 6, wherein, when the load is a third predetermined load applied to the pedal pad, the pedal arm further contacts a different engagement position of contact of the engagement surface than the engagement position of contact when the load is the second predetermined load to generate a third return force on the pedal pad, the third predetermined load is a greater load than the second predetermined load, and the third return force is a greater force than the second return force.

8. The pedal emulator assembly of claim 1, wherein the lower arm is fixedly coupled to within the cavity of the housing.

9. The pedal emulator assembly of claim 1, wherein the lower arm further includes a concave portion configured to receive a pivot portion of a pivot end of the pedal arm, such that the pedal arm pivots about the pivot portion within the concave portion of the lower arm.

10. The pedal emulator assembly of claim 1, wherein the compressible member includes a plurality of springs coaxially aligned about the spring carrier.

11. The pedal emulator assembly of claim 1, further comprising: a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip; wherein the sensor assembly is configured to detect movement of the pedal arm.

12. A pedal emulator assembly for a vehicle comprising: a floor mounted housing having a pair of sidewalls, a first end wall, and a second end wall; a lower arm having a spring retaining portion formed therein; a pedal arm having a pair of opposing walls, an interior surface, and a pedal pad, the pedal arm pivotally coupled the housing such that the pair of opposing walls are movable relative to the pair of sidewalls thereby defining a cavity by the pair of opposing walls and the

housing; a lever arm pivotally coupled to the housing, the lever arm further comprising: a first end positioned at a distal end of the lever arm; a second end positioned opposite the first end; a hub portion disposed at the second end; an upper frame portion and a lower frame portion that connect at the distal end and at the hub portion, the upper frame portion having an engagement surface configured to slidably engage with a portion of the interior surface of the pedal arm, the engagement surface having a non-linear profile that extends a length between the distal end of the lever arm and the hub portion, the non-linear profile configured to provide a different force characteristic based on an engagement position of contact between the portion of the pedal arm and the engagement surface substantially along the length of the non-linear profile based on a load applied to the pedal pad, the engagement surface is positioned to be enclosed within the cavity; and a spring receiving portion positioned between the first end and the second end and within the lower frame portion; and a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion, wherein, when the load is a first predetermined load applied to the pedal pad, the interior surface of the pedal arm drives the lever arm into the compressible member such that the compressible member compresses into a compressed state to generate a first return force on the pedal pad.

13. The pedal emulator assembly of claim 12, wherein the interior surface of the pedal arm is non-linear.

14. The pedal emulator assembly of claim 12, wherein the upper frame portion includes a first portion extending between the first end of the lever arm and an apex of the upper frame portion and a second portion extending between the apex of the upper frame portion and the second end of the lever arm.

15. The pedal emulator assembly of claim 14, wherein the engagement surface of the first portion of the upper frame portion has a first slope and the engagement surface of the second portion of the upper frame portion has a second slope different from the first slope.

16. The pedal emulator assembly of claim 12, wherein, when the load is a second predetermined load applied to the pedal pad, the interior surface of the pedal arm contacts a different engagement position of contact of the engagement surface than the engagement position of contact when the load is the first predetermined load to generate a second return force on the pedal pad, the second predetermined load is a greater load than the first predetermined load, and the second return force is a greater force than the first return force.

17. The pedal emulator assembly of claim 16, wherein, when the load is a third predetermined load applied to the pedal pad, the interior surface of the pedal arm contacts a different engagement position of contact of the engagement surface than the engagement position of contact when the load is the second predetermined load to generate a third return force on the pedal pad, the third predetermined load is a greater load than the second predetermined load, and the third return force is a greater force than the second return force.

18. The pedal emulator assembly of claim 12, further comprising: a sensor assembly positioned within the housing, the sensor assembly further comprising: a circuit board; at least one Hall Effect chip communicatively coupled to the circuit board; and a magnet communicatively coupled to the at least one Hall Effect chip, wherein the sensor assembly is configured to detect movement of the pedal arm.

19. A pedal emulator assembly for a vehicle comprising: a housing defined by a pair of sidewalls, a first end wall and a second end wall; a lower arm having a spring retaining portion formed therein; a pedal arm having a pair of opposing walls, a pivot portion having a pivot end pivotally coupled to the housing, a pedal pad, and an interior surface that is non-linear, the pedal arm pivotally coupled the housing such that the pair of opposing walls are movable relative to the pair of sidewalls thereby defining a cavity by the pair of opposing walls and the housing; a lever arm positioned within the cavity of the housing and pivotally coupled to the housing, the lever arm further comprising: a first end positioned at a distal end of the lever arm; a second end positioned opposite

the first end; a hub portion disposed at the second end; an upper frame and a lower frame that connect at an apex at the distal end and at the hub portion, the upper frame including an engagement surface with a non-linear profile configured to slidably engage with a portion of the interior surface of the pedal arm, the engagement surface is positioned to be enclosed within the cavity; and a spring receiving portion positioned between the first end and the second end in the lower frame; and a spring carrier including a compressible member extending between the spring receiving portion and the spring retaining portion, wherein, when a first predetermined load is applied to the pedal pad, the interior surface of the pedal arm contacts the engagement surface of the upper frame of the lever arm, such that the lever arm is forced into the compressible member to generate a return force on the pedal pad.

20. The pedal emulator assembly of claim 1, wherein the engagement surface extending between the first end and the second end such that the length of the engagement surface is greater than half of a distance of the lever arm between the first end and the second end.
